Notes on Thermodynamics and Related

Burak Ozsoy

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Abstract

This paper is not a textbook, and it is not trying to be one. It is closer to a production of interest: a set of notes on the principles, methods, and details of thermodynamics that I found exciting but that, in my experience at least, engineering classes tend to walk past. The topics here are all related, but I was not aiming for the completeness or sequencing of a proper course. I wanted instead to give the reader a chance to see things that are rarely taught, and to relay some of what makes thermodynamics, to me, one of the most phenomenal subjects in physics: built from airtight reasoning, and yet quietly governing every other field in physics and engineering, whether or not anyone points that out. I wanted to convey the excitement of watching thermodynamics evolve, from a simple thought experiment about a reversible engine, all the way to predicting the eventual death of the universe, and further still into engineering marvels like density functional theory and modern thermodynamic simulations. I did not manage to go as deep into the physics as I would have liked, and I never reach, for example, Boltzmann, partly because I assumed background knowledge many readers would not have, and partly because I would rather leave the reader room to go discover that on their own. I am a student writing this, not an authority, so I offer it in that spirit. I hope you come to like thermodynamics as much as I do, and I want you to know that there are a billion more interesting things related to engineering and physics in this subject than I was able to fit here. To see that for yourself, you need do no more than open the table of contents of any serious analytical thermodynamics textbook (S.L. Loo's, for example, I particularly liked) and let your imagination go, or really, just look at almost anything. For a sense of the specific ground covered in this paper, from Carnot's original reasoning to thermodynamic potentials, diffusion, and thermoelectric devices, see the table of contents below.

Contents

1 Everything is Thermodynamics!

Here is a strange fact to sit with for a moment. The law that predicts how fast your coffee cools on the counter is, without modification, without a single term changed, the same law that predicts the ultimate fate of the entire universe, whether it ends in ice or in fire, trillions of years from now. A cooling mug and the death of everything are, physically speaking, the same problem. That should surprise you. Almost nothing else in physics can make that claim. The rules governing a hydrogen atom say nothing useful about a hurricane. The rules governing a hurricane say nothing useful about a black hole. But the rules governing your coffee cup, it turns out, scale all the way up, and it's worth asking why, before you've seen a single derivation.

The answer is that thermodynamics doesn't describe what things are made of. Every other branch of physics you'll ever study earns its power by getting specific: quantum mechanics needs to know it's dealing with electrons and photons, general relativity needs to know it's dealing with mass and curvature, fluid dynamics needs viscosity and density. Thermodynamics asks for none of that. It doesn't care whether the system in front of you is a gas of argon atoms, a magnet, a rubber band, a neuron, or a star. It asks only one kind of question: given what this system is allowed to exchange with its surroundings, energy, particles, volume, what is it forbidden from doing? And it turns out that question has a complete, exact, and shockingly short answer, one that was finished, essentially, by 1865. Most of physics is a theory of what happens. Thermodynamics is a theory of what cannot, and there is a peculiar kind of power in being able to say "never" with total confidence about a system you've never even looked at.

Watch how far that single kind of confidence actually reaches. It reaches into machines: every engine ever built, from Newcomen's coal-fired pump to the fusion reactor someone is trying to switch on right now, is bounded by an efficiency ceiling that a French army engineer worked out in 1824 using nothing but a piston and pure logic, a ceiling that depends on two temperatures and nothing else, and that no engineering effort has ever beaten or ever will. It reaches into chemistry: every reaction that has ever run to completion, rusting iron, burning gasoline, a neuron firing, a virus assembling itself inside a cell, did so because it was headed toward a state of lower free energy, and no reaction in the history of the universe has ever run the other way on its own. It reaches into biology in a way that should genuinely unsettle you: a living organism is, for as long as it's alive, a small pocket of stunning order, one of the most improbable arrangements of atoms you will ever encounter, and it is allowed to exist in open defiance of the universe's overwhelming preference for disorder only because it is constantly paying rent, in heat, out into its surroundings, to stay that way. Stop paying, and the disorder it was holding at bay arrives immediately. That is, in the most literal thermodynamic sense available, what it means to die.

And it reaches further than matter itself. In 1961, a physicist named Rolf Landauer proved that erasing a single bit of information, not storing it, not moving it, just erasing it, has an unavoidable minimum energy cost, set by the same constant that governs steam engines. Delete enough data, anywhere, on any hardware humanity ever builds, and you are paying a thermodynamic tax that was fixed before computers existed. It reaches into astrophysics: a black hole, an object made of nothing but warped empty space with no atoms in it at all, has a temperature and an entropy, computed not by chemistry but by the geometry of its event horizon, obeying the exact same second law that governs your coffee cup. Stephen Hawking spent years resisting his own proof of this before accepting it, because it meant that even an object built from pure gravity has to answer to thermodynamics. Nothing gets an exemption.

That is the actual claim buried inside a title as sweeping as "Everything is Thermodynamics." Not that thermodynamics is a useful tool worth learning, though it is, but that a set of constraints discovered by watching steam escape a piston in the early nineteenth century turned out, without anyone intending it, to be a law of the universe on par with anything Newton or Einstein wrote down, one that governs your kitchen, your laptop, every living cell in your body, and the eventual fate of every star in the sky, all with the same handful of inequalities.

This book exists to earn that claim rather than assert it. We will not start from the end result and ask you to trust it. We will start, deliberately, from almost nothing, two refusals so mundane you've probably never doubted either of them, and build forward from there, one forced logical step at a time, watching entropy, free energy, and everything downstream of them arrive not as inventions but as inevitabilities. By the time you reach the last chapter, the sentence "everything is thermodynamics" should no longer read like a slogan. It should read like something you already knew was true, and were simply waiting to be shown why.

2 State Functions & Importance

The quantities whose values determine the state of a system are called its thermodynamic coordinates ([6]). Suppose you have a metal block with 10 joules of energy and you put it through all kinds of processes. You throw it into the furnace and heat it, cool it in Antarctica, bring it down to Mariana trench, take it to the moon and back, if after all of that the block ends up with the same internal energy, volume, and number of particles then thermodynamically it is in the same state as before. The metal block's history doesn't matter, only its current thermodynamic coordinates matter.

> A thermodynamic state is determined completely by its state variables; it is independent of the process by which the system arrived there.

A state function is a function of thermodynamic coordinates. They are either defined by other state variables like volume, mass, pressure, internal energy, enthalpy, temperature, number of particles,...etc.

$$H(U,P,V) = U{+}PV \quad \text{(Enthalpy)}, \qquad \rho(m,V) = \frac{m}{V}, \qquad G(p,T) = H{-}TS \quad \text{(Gibbs Free Energy)}$$

or they are constrained by other state variables for a given thermodynamic equilibrium.

$$PV = nRT \;\; \text{(For Ideal Gasses)}$$

> A state function F is a quantity that depends only on the state variables at a given state of thermodynamic equilibrium, and is independent from the path taken to reach the state.

Important 1 - Degrees of Freedom: A thermodynamic system has a fixed number of independent degrees of freedom. Once a complete set of independent state variables is specified, all other state variables are uniquely determined through the system's equations of state and constitutive relations. They cannot take arbitrary values. Their values are constrained by the chosen coordinates and the physics of the system. For instance, if P and V are assigned, T is uniquely determined for the given n and R.

Important 2 (Dr.Lage's) - Mathematical Notations:

Definitions in thermodynamics are very important and very specific. We have to be such with the math as well. State functions and path-dependent variables are defined as

$$\begin{aligned} dX &: \text{ exact differential (state function)} \\ \delta X &: \text{ inexact differential (path-dependent quantity)} \end{aligned}$$

For state functions

$$\oint dU = 0, \qquad \oint dS = 0, \qquad \oint dP = 0,$$

Because if the state starts at a point, for example at U_i, and takes some arbitrary path and ends back at the same state, $U_f = U_i$, then we would have no way of knowing as the state is defined the same as earlier. However for path dependent variables,

$$\oint \delta W \neq 0, \qquad \oint \delta Q \neq 0,$$

(unless a special circumstance is arranged such that there is a unique path) because it doesn't make sense to talk about a Q_f, Q_i, W_f or W_f. Heat is not a thermodynamic coordinate, it is a process which by definition is a change in the thermodynamic coordinates [6]. Same is true for work. δQ and δW are therefore inexact differentials. Can you think of an "initial" work state (and don't try to think of initial work *done* minus the final work *done*, that would be the net work *done*) ?

But you might object, what about the fact that

$$dU = \delta Q - \delta W$$

The difference $\delta Q - \delta W$ is path independent even though the constituents are not. This is the first law of thermodynamics.

Important 3 – Comparing: Be careful in your definitions. If you don't understand the difference between the same internal energy and the same thermodynamic state, for example, you might be inclined to suggest a metal block with 10 joules is in the same thermodynamic state as your uncle in Siberia who also has 10 joules. Even though internal energy is a state function, it is not sufficient to define the state without the other thermodynamic coordinates. Your uncle and the metal block are not in the same thermodynamic state due to number of particle, pressure, temperature, and other state variables in concern.

2.1 Mathematical Realizations

: If we have a function $F(T, V)$, we can express as small differential change as

$$dF = \left(\frac{\partial F}{\partial T}\right)_V dT + \left(\frac{\partial F}{\partial V}\right)_T dV$$
$$dF = MdT + NdV$$

Thus we require, innately, that

$$\frac{\partial M}{\partial V} = \frac{\partial N}{\partial T}$$

For state functions this is condition holds, it doesn't for path dependent variables like heat.

But why is this important? Recognize that if I decided to replace F with the internal energy U (in fact that's why I defined F as $F(T,V)$, to make it consistent with the differential for U) and then we took the closed loop integral of dU

$$\oint dU = \oint MdT + NdV = 0$$

This is a conservative vector field. This is a good place to remind you of Stokes Theorem:

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

The equality condition written previously above is the curl free expression that you might establish an analogy between with electromagnetism as

$$\vec{\nabla} \times \vec{E} = 0 \rightleftharpoons \frac{\partial M}{\partial V} = \frac{\partial N}{\partial T}$$

This is very important as it allows us to map the entire space as a scalar potential field, which we can literally tabulate (just like electric potential ϕ). **This is why we can have charts and tables for state functions, regardless of material, at the end of our textbooks!!!**

3 Reversible and Irreversible Processes

> **Definition:** A process is any change in the thermodynamic coordinates of a system [6].

3.1 Reversible Processes:

If I could change something, for example boil a metal, and somehow reverse this change by bringing the metal back to its previous form without changing anything in the universe, then the processes would be reversible.

> **Definition:** Reversible processes are ideal processes that can be reversed in such a way that both the system and its surroundings are returned exactly to their original states with no net change anywhere.

In the explanation of reversible processes you likely come across examples such as "a reversible process is a succession of equilibrium states for example heating a water very slowly so that at each point in the heating process the system is infinitesimally close to equilibrium". Now what does that mean and why is the succession of equilibrium states important? And say we compress a gas very fast or very slowly, the initial and the final states are still the same so what is different?

First, the initial and final states are defined by state functions that are independent of path, however we are talking about the qualities of the processes. So how we get there matters.

Second, in a process, if the state of the system is always infinitesimally away from equilibrium then there is an infinitesimal net driving force (e.g. temperature gradient $\rightarrow$ heating) that would cause an infinitesimal imbalance between the system and the surrounding. For example, if a gas is slowly compressed and it is always infinitesimally near equilibrium, what does that mean? It means that there is an infinitesimally small reason for there to be an imbalance in pressure, temperature,...etc. between the system and surrounding. Thus if we changed the pressure of the gas in an ideal frictionless piston by compressing, $P + dP$, then we can easily let our hand go and the gas would expand by $-dP$ perfectly reversing the situation.

Of a piston, if you are thinking of the system at its initial state, and then at its final with higher pressure and higher temperature, you are thinking wrong. Remember, we never said that there is zero exchange with the surrounding. At each step, the system differs infinitesimally from the surroundings, creating a small driving force. Because of these infinitesimal imbalances, applying an opposite infinitesimal change $-\epsilon$ will return the system to its previous state. Repeating this step by step allows the system to retrace its entire path back to the initial state, while exchanging exactly the same amounts of heat and work with the surroundings in reverse.

Example: We are compressing a piston by dP each time. What is the work if the process is reversible? Work is defined as

$$\delta W = P_{\text{ext}} dV$$

By our definition change in the pressure is

$$P_{\text{ext}} = P_{\text{system}} + dP, \quad dP \rightarrow 0$$

Thus

$$W = \int_{V_i}^{V_f} P_{\text{sys}} dV$$

If it were irreversible we would have the following instead

$$W = \int_{V_i}^{V_f} P_{\text{ext}} dV$$

3.2 Irreversible

Almost all processes that we see, however, are irreversible processes. Some processes really close to reversible processes are very small currents through a straight wire or orbital motion of planets for example. However, even they produce heat, tides, and radiation that technically deem these processes irreversible.

Once the a process starts, the system and the surrounding becomes coupled and starts to exchange information (exchange energy, mass or momentum and can be in the form of work, heat or mass flow for example). The configuration of the system and the surrounding change, and the thermodynamic coordinates in space change. Then you reach a final state and want to retrace your steps but what you realize is that the details of the microscopic state of the system have become correlated with an enormous number of environmental degrees of freedom.

For example, for a piston, you can control the pressure of the gas and the piston position, but can you control the temperature gradients in the wall, vibrations you create, sound waves? In real life there is no infinitesimal change. Change always happen in finite amount and these changes and their consequence can not be re-traced.

Another example can be having a box that is divided into two, and in one side there is particles. When you lift the barrier those particles will diffuse and start bumping into the walls imparting momentum, creating a temperature profile within the box, causing vibrational waves in the box,..etc.. If you somehow put all the particles back to their original place without doing work (adding or subtracting energy from the surrounding), macroscopic remnants of correlations remain (e.g. temperature fields, vibrations), which retain partial information about the past state. So the information spread and is now dispersed into multiple degrees of freedom.

Even though, in principle, the laws of mechanics are time reversible even in the microscopic level, if one could prepare the system and surroundings in exactly the time reversed microscopic state, the process would run backward. However, in reality, the system evolves into a vastly larger set of microstates, with a vast number of correlations (inferable information of A from B) dispersed throughout the surroundings making the required reversed initial condition physically unattainable in practice and effectively impossible for macroscopic systems.

At this point, I might have lost you so I will briefly re-state that if there is a change in the system, you can restore the system back into its original state, however the thermodynamic coordinates in the universe has changed. This is called an irreversible process.

3.3 Clausius, Kelvin and Carnot Went to a Bar in Austria

Unfortunately due to the time this paper took I can not go over this crucial chapter which is very misunderstood and very undervalued in the detail I would be satisfied with. If you are an engineer don't read but if you are curious I would suggest you dig deeper into these.

1. Clausius states,

   > "It is impossible to construct an engine that, operating continuously [cyclically], will produce no effect other than transfer of heat from cooler to hotter body" (S.L.Loo)

   This statement is not merely an obvious observation but a constraint on all thermodynamic cycles. Clausius figured it through a careful study of Carnot's theory of reversible heat engines. He realized that Carnot's efficiency argument implicitly requires that spontaneous heat transfer cannot occur from cold to hot without external work. Otherwise, one could construct a cyclic device that produces net work without any other effect, violating the structure of thermodynamic cycles. Thus, Clausius' statement encodes the fundamental directionality of thermodynamic processes. It does not claim that a single state has an intrinsic direction of heat flow, but rather that transitions between states are constrained in direction.

2. Now you will figure Carnot's brilliance. Figure how reversibility implies Carnot to deduce that the efficiency of an engine only depends on temperature differences.

3. Figure how Clausius's statement is equivalent to Kelvin-Planck statement,

   > "No process is possible whose sole result is the abstraction of heat from a single reservoir and the performance of an equivalent amount of work" [6]

3.4 Carnot Discovers Water

Sadi Carnot was born into an era of great revolutions amid which, perhaps, he led one of the greatest whose impact will live on forever in human history [65]. His father, Lazare Carnot, was one of Napoleon Bonaparte's generals and also a passionate scientist that contributed greatly to the field of thermodynamics. In fact, he published a book in 1803 called "Fundamental principles on equilibrium and movement" in which he defended the impossibility of perpetual motion [2].

Thus, Sadi Carnot became influenced and drawn into his father's study and, inadvertently, the greatest technological development of their era, the steam engine, through discussions with his father Lazare. Sadi quickly recognized the significance of this machine and understood the crucial role it would play in determining the dominance and survival of the European powers. As a committed French nationalist concerned with France's standing, and realizing that his country was falling behind in steam engine design, he set out to improve steam engines [3][4]. He discovered whether there was a limit on how much one can improve the efficiency of the steam engine. **He found an upper bound in thermodynamic efficiency**.

This idea, this discovery, must sound familiar. The fact that an upper bound exists suggests an important truth about our reality, a law is broken if one goes over the limit. Sadi pondered this in his paper

> "...whether... a limit which the nature of things will not allow to be passed by any means whatever; or on the contrary,..." [3]

But now that we know there is a limit, which law is this that we would break? Certainly not the conservation of energy because it alone does allow an efficiency of 100% which is not the case. But which law then? Which underlying law protects and implies this Carnot efficiency and prevents us from being more efficient?

Among other engineering ingenuity and brilliance (such as Carnotization which we use in modern power generators) Carnot showed us in his "Reflexions sur la puissance motrice du feu et sur les machines propres a developper cette puissance", I believe this question he left to us was the biggest, paving the way to the development of the second law of thermodynamics.

Note: The reader should not attempt to read Carnot's paper with a preconceived notion that it is flawless. It has many bad arguments and false conclusions as well, mostly stemming out of his belief in the caloric theory. Interestingly though, he does show signs of doubt in the caloric theory, and in his later notes, he outright rejects caloric theory for mechanical theory of heat.

3.4.1 On the Production of Motion: Brilliance 1 [3]

Carnot's first major insight was recognizing that the motive power (mechanical work) produced by a heat engine arises not from the consumption of caloric, nor from the expansive force of steam, but from the passage of heat from a higher temperature to a lower temperature. The working fluid (steam in the case of the steam engine), he states, serves only as a transport medium.

To reach this conclusion, Carnot analyzed the sequence of operations by which a steam engine functions:

1. Heat generated by combustion in the furnace passes through the walls of the boiler.
2. This heat vaporizes the water, producing steam at high temperature and pressure.
3. The steam is admitted into the cylinder, performing work on the piston.
4. The steam is then directed to the condenser, where it is condensed.
5. The condensate (liquid water) is pumped back to the boiler, completing the cycle.

He observes the "re-establishing of equilibrium in the caloric", that is the passage of heat. From this operational description, Carnot makes the conceptual leap, he asks: Would the engine continue to produce work if the cold reservoir, the condenser, were removed? The answer is clearly no. Without the condenser, the temperature of the exhaust steam could never be reduced, no temperature difference would exist to impel heat flow. Without such a difference, there would be no transfer of heat, and Carnot notes that if heat drives motion then without it no mechanical work can ever be produced.

Thus, Carnot concludes that the essential condition for the production of motive power is the passage of heat from a body at higher temperature to one at lower temperature. When this passage stops, the engine ceases to function. The motive of power of an arbitrary engine, fundamentally, is the difference between the hot and the cold reservoirs. Additionally, he states the reciprocal is true as well: work can produce a temperature difference. These are the seed of the second law of thermo-dynamics.

You should be tempted to ask "what about Newtons heat convection equation and Fourier's conduction heat transfer law, shouldn't these render Carnot's conclusion trivial ¿' The answer is no. Remember, back then heat was not well understood and what produced work, motive of power, was a big question. Fourier developed his heat conduction theory independent of any resolution of what heat actually was.

3.4.2 Advantage of the working fluid?: Brilliance 2 & 3

An important question at the time was whether the amount of work that could be generated between two temperature reservoirs depended on the intermediary agent, or working substance, such as steam, oil, cherry juice,...etc. To answer this question, Carnot introduced the great idea of reversibility. He suggested that just as steam can produce work by driving a piston, work can also be done on the piston to cool and expand (rarefy) the steam inside it. This was a remarkable insight as it contained the seeds of the modern concept of conservation of energy, or as we know it, the First Law of Thermodynamics, even though this principle had not yet been formulated.

Carnot then argued that every working substance should produce the same efficiency when operating between the same two temperature reservoirs. Now he asks, if there is a superior working substance capable of increasing efficiency, and we use it to build a Super Carnot engine. If we connect this Super Carnot engine to an inverse ordinary Carnot engine but with an inferior regular working substance, this would surely produce a perpetual machine generating unlimited work without any net expenditure of energy. This contradiction arises because a reversible engine is, by definition, an engine free from internal sources of loss such as friction. In contrast, the reduced efficiency of real engines, such as those found in automobiles, is caused by irreversible processes including friction, heat transfer across finite temperature differences, combustion imperfections, and other dissipative effects. These irreversible processes prevent real engines from achieving the ideal performance of a reversible engine. I will discuss the proof of this argument in much greater detail in the next section.

However, you might still be tempted to compare something like honey and water, or air for example. Of course, the performance of the engine will drop significantly with honey for instance, and Carnot doesn't suggest otherwise. What is wrong in comparing honey to water as a working fluid is that even though the friction between piston, the cylinders, and the honey is negligible, the viscosity of the honey will produce internal viscous effects causing internal friction. So the reason you chose honey to compare with (that is conspicuously viscous) is the reason such a question fails, in the first place.

3.4.3 Limit on Efficiency

Carnot's efficiency argument stands upon three pillars:

1. Heat engines work by transferring caloric from a hotter body to a colder one.
2. The most efficient engine is an engine with no irreversible process (a reversible engine).
3. All reversible engines operating between the same reservoirs have the same efficiency regardless of the working fluid.

Now, the 3rd we briefly proved in the previous section, so I will rigorously prove it here.

Proof:

Assume we have two reversible engines operating between the same reservoirs, however, with two different efficiencies e_1 for the Carnot engine and for the super-Carnot engine e_2. Carnot states that for every reversible engine, we can run that engine in reverse like a refrigerator (this is embedded in the definition of reversibility as Carnot defined, just reverse the arrows on the P-V graph). Thus we will construct a system where the Carnot engine will feed work into the super-Carnot engine that will use that work as a refrigerant.

The Carnot engine and the super-Carnot engine are connected to the same two reservoir. The Carnot engine will take in Q_1 from the hot reservoir, produce W_C and reject $Q_1 - W_C$ to the cold reservoir.

On the refrigeration side, since the super-Carnot engine is more efficient by a factor of $1+\chi$ than the Carnot engine, super-Carnot refrigerator will use the provided W_C to extract $Q_1(1+\chi)$ from the cold reservoir and reject $Q_1(1+\chi)+W_C$ to the hot reservoir.

Now let's see the overall effect of these two working together:

$$\begin{aligned} \text{From the cold reservoir:} \quad & Q_1(1+\chi) - (Q_1 - W_c) \\ \text{To the hot reservoir:} \quad & \big(Q_1(1+\chi) + W_c\big) - Q_1 \\ \text{Net work:} \quad & 0 \end{aligned}$$

We have just created a perpetual refrigeration engine that would run forever with zero work input. This is impossible. By logic, since the second statement above is true, then by what we have just showed, reversible engines operating between two identical reservoirs must have identical efficiencies, or else they create perpetual machine of the second kind.

Now, let get back on track. So, since we established that all reversible engines between the same two temperatures must produce the same amount of work per unit caloric, Carnot concludes that, therefore, the amount of work per unit caloric is a universal function of the temperatures, because as we saw, work output only depends on reservoir temperatures and nothing else.

$$\frac{W_{\text{net}}}{Q_{\text{total}}} \propto f(T_{\text{hot}}, T_{\text{cold}})$$

This is the great leap. Here we have concluded that the efficiency is only a function of the assigned temperatures of the reservoirs from the fact that all reversible engines operating between the same reservoirs must produce the same work per caloric. This function is not only universal but it sets a limit to the performance, depending on nothing else but the reservoirs being operated between. Carnot then effectively posits that a reversible engine gives motive power proportional to the caloric it receives, then the same engine, running backward, consumes motive power proportional to the caloric it restores. Thus, the caloric gained and rejected must be in a constant ratio that depends only on the temperatures of the reservoirs. Mathematically

$$\text{Carnot Engine:} \qquad W = Q_{\text{hot}} \cdot f(T_{\text{hot}}, T_{\text{cold}})$$

$$\text{Reverse Carnot Engine:} \qquad W = Q_{\text{cold}} \cdot f(T_{\text{cold}}, T_{\text{hot}})$$

Since together in a loop they should yield nothing with equal work consumed and produced,

$$Q_{\text{cold}} \cdot f(T_{\text{cold}}, T_{\text{hot}}) = Q_{\text{hot}} \cdot f(T_{\text{hot}}, T_{\text{cold}})$$

$$\frac{Q_{\text{cold}}}{Q_{\text{hot}}} = \frac{f(T_{\text{cold}}, T_{\text{hot}})}{f(T_{\text{hot}}, T_{\text{cold}})}$$

I tried to stay away as much algebra as possible since it was actually Clapeyron, Lord Kelvin and Clausius who developed Carnot's ideas and turned them into mathematical formalizations. Carnot doesn't explicitly write out the Carnot efficiency as we know it today, for example. His paper while including some calculations, contributes mainly by the logical arguments he develops.

Lastly, I want to express something quite interesting and very ahead that Carnot figured as Herman Erlichson noticed [5]. Using the conservation of energy law we can obtain

$$W = Q_{\text{hot}} - Q_{\text{cold}} = Q_{\text{hot}} \left(1 - \frac{Q_{\text{cold}}}{Q_{\text{hot}}}\right)$$

And from Lord Kelvins work

$$\frac{Q_{\text{hot}}}{Q_{\text{cold}}} = \frac{T_{\text{hot}}}{T_{\text{cold}}}$$

Thus we can produce

$$W = Q_{\text{hot}} \left(1 - \frac{T_{\text{cold}}}{T_{\text{hot}}}\right) = Q_{\text{hot}} \left(\frac{T_{\text{hot}} - T_{\text{cold}}}{T_{\text{hot}}}\right)$$

From Carnot, we know that the motive power increases with temperature difference and he further adds in his paper that "the fall of caloric produces more motive power at inferior than at superior temperatures". This statement implies that the work produced cannot depend solely on the temperature difference between the reservoirs. To see this, consider two processes with the same temperature drop,

$$400^\circ\text{C} \rightarrow 390^\circ\text{C} \qquad \text{and} \qquad 100^\circ\text{C} \rightarrow 90^\circ\text{C}$$

both of which have $\Delta T = 10^\circ$C. If the work depended only on ΔT, these two processes would produce the same amount of work. However, Carnot explicitly argues that the latter process yields more motive power. Therefore, the work can not depend only on the temperature difference. A function that satisfies his logic is unsurprisingly the following

$$W \propto Q_{\text{hot}} \left(\frac{T_{\text{hot}} - T_{\text{cold}}}{T_{\text{hot}}} \right)$$

This is huge! Of course, Carnot didn't derive the above formulation as he believed in the caloric theory ($Q_h = Q_c$), but he correctly observed the functional behavior this universal efficiency function must depend upon. Thus his intuition accurately points towards $f(T_{\text{hot}}, T_{\text{hot}})$ and not $f(\Delta T)$, which is discerned without conservation of energy or Lord Kelvin's work, just using logic and insight. It's brilliant.

3.5 Kelvin Invented Draft Beer

William Thomson, later known as Lord Kelvin, is one of the giants in the history of physics, particularly in the development of thermodynamics. He studied under the renowned French experimentalist Victor Regnault and later collaborated extensively with James Prescott Joule to develop the Joule-Thomson effect and mechanical theory of heat [2]. His role in our journey toward understanding entropy was to place Carnot's ideas on a firmer theoretical and mathematical foundation and to introduce an absolute temperature scale that is independent of the properties of any thermometric substance [6]. This reformulation placed Carnot's theory on a universal footing and ultimately paved the way for the modern definition of thermodynamic efficiency and, later, the concept of entropy.

We take the non-relative, absolute temperature scale Kelvin invented for granted these days, and have forgotten the grand implications his work paved the way to. An absolute temperature scale is a paradoxical invention since devising a thermometry standard that is not relative to the thermometric properties of the substance used, has to use a substance regardless with a coefficient of expansion that is a function of temperature. This means to measure the temperature of something using a substance without depending on that substance. However, this was an important task because a major issue back then was the inability of consistent temperature measurements as different thermometers did not agree with one another, especially away from the calibration points. Intervals changed due to different substance expansion reactances and circumstances. Thus, Kelvin set out to define temperature without using any material which was a huge jump of thought since temperature back then was seen as an empirical variable. Before Kelvin, temperature was operationally defined through the behavior of thermometric substances. Kelvin reversed this viewpoint by defining temperature from thermodynamic principles, allowing thermometers to become instruments that measure an already-defined quantity. So next time you look at a thermometer keep in mind Kelvin's audacity and courage to take on a challenge that is both paradoxical, and ill-defined.

Before we begin you might still ask, "Why did we use just a thermometer like Regnault's precise and operational air thermometry? In our daily lives we use Fahrenheit and Celsius scales and so far we never missed them, why should we seek this 'absolute Kelvin scale'? What is the difference between the definition given by the Celsius scale and the Kelvin scale?" This is a fundamental question. Because even when you think about the equations we use in thermodynamics like the ideal gas law,

$$PV = nRT$$

instead of using Kelvins for T, we can simply change the units of R.

Thermometers use a substance that expand with temperature. How much the temperature change is per expansion is something we literally make up. For example, we can say ice is the zeroth degree, and every centimeter the mercury expands is equal to 1 Jack degrees, we just invented the Jack scale. But as said, different substances expand differently and often this expansion is a nonlinear function of the temperature. Regnault solved the practical problem of measuring temperature consistently. Kelvin addressed a deeper conceptual question. In an analogy, Regnault improved the ruler but Kelvin defined what the ruler measures.

What Kelvin realized is totally different. Kelvin sought to define temperature as a physical quantity determined by universal thermodynamic laws rather than by the behavior of a particular thermometric substance. Carnot had already shown that every reversible heat engine shares a universal efficiency independent of its working substance. Kelvin recognized that this universality must reflect a property of nature itself rather than of steam, air, or mercury. So, if engine efficiency depends only on temperature, perhaps temperature itself should be defined through that universal law. Thus, now we know that 0 K corresponds to absolute zero, the point at which no lower thermodynamic temperature is possible, whereas 0 C is arbitrary human convention (arbitrarily chosen to correspond to the temperature of ice, it could've equally been chosen to be the freezing point of tritium if it was as accessible and useful). Another example is while 200K/400K means half the thermodynamic temperature, 200C/400C doesn't mean anything.

In the following sections, I rely heavily on *The Absolute and Its Measurement: William Thomson on Temperature* by Chang and Yi [7]. While their paper provides an excellent historical and technical account of Kelvin's development of the absolute temperature scale, some of the arguments can be difficult for readers encountering the subject for the first time. Accordingly, I have attempted to simplify and expand upon several of their explanations where I felt additional intuition would benefit beginners. Readers seeking a more comprehensive historical discussion of Kelvin's work on absolute temperature, are encouraged to consult the original paper.

3.5.1 Kelvin's First Absolute Temperature Scale

Kelvin recognized that Carnot's theory provided the first universal relationship between heat, work, and temperature that was independent of the particular working substance. If every reversible heat engine possesses the same efficiency regardless of whether it uses steam, air, or any other fluid, then perhaps temperature itself should be defined through this universal property rather than through the expansion of a thermometer. His first proposal was therefore to construct a temperature scale such that equal intervals of temperature corresponded to equal amounts of mechanical work obtainable from a given quantity of heat transferred through a reversible engine.

To give this definition practical meaning, Kelvin considered an ideal Carnot engine operating between two heat reservoirs: one maintained at a fixed reference temperature and the other at the temperature to be assigned on the new scale. The work performed by the engine would then determine the temperature difference between the reservoirs. Thus, equal temperature intervals would correspond to equal amounts of work obtainable from one unit of heat.

$$\frac{W}{H} = (T_{\text{Hot}} - T_{\text{Cold}})\mu$$

Unfortunately, this definition could not be realized directly because a perfectly reversible Carnot engine is an idealization, it just doesn't exist and it is not practical. Instead, Kelvin estimated the behavior of such an engine using the most accurate experimental data available. He relied primarily on Henri Regnault's precise measurements of the properties of saturated steam, which allowed him to approximate the work that an ideal reversible engine would produce. In modern notation, this work can be expressed as

$$W = \int_{p_1}^{p_2} \psi \, dp$$

where ψ denotes the difference in volume between the expansion and compression curves of the reversible cycle at a given pressure. To estimate the work produced by the reversible cycle, Kelvin needed to determine the change in volume due to the vaporization of the water by heating. According to the caloric theory, a quantity of heat H (treated as a conserved fluid-like substance) supplied to water produced a corresponding mass of steam through the latent heat of vaporization. Since the work depends on the increase in volume during vaporization, Kelvin considered the difference between the volume occupied by the steam and the original volume of the liquid water. Using the ratio σ of the specific volumes (or equivalently densities) of water and steam, this volume change becomes proportional to (1- σ). Quantities like latent heat required to produce a unit volume of saturated steam at a given temperature (k) and ratio of density of steam to density of water at a given temperature (σ) were tabulated pretty well back then thus Kelvin used them to formulate

$$W = \int_{p_1}^{p_2} (1 - \sigma) \frac{H}{k} dp$$

Given this cycle operates between temperature reservoir T_H and T_C, and because all of the pre-mentioned variables are measured as a function of the air thermometer temperature T except H, Kelvin re-wrote the above as

$$\frac{W}{H} = \int_{T_H}^{T_C} \frac{(1 - \sigma)}{k} \frac{dp}{dT} dT$$

Using Regnault's extensive tables for k, σ, and the saturation pressure as functions of the air-thermometer temperature, Kelvin evaluated the integral numerically and thereby determined $\frac{W}{H}$ as a function of air-thermometer readings. Although Kelvin had succeeded in expressing W/H in terms of experimentally measurable quantities, the definition remained difficult to apply in practice because it relied on approximating the behavior of an ideal reversible engine. But more importantly, the caloric theory on which the derivation rested was soon undermined by Joule's demonstration that heat is not a conserved substance but a form of energy. This forced Kelvin to reconsider the caloric framework he and Carnot had so far relied on.

3.5.2 Quick Interlude: Why the Failure of Caloric Theory Prompted Re-Formulation

I added this section because I was worried I wasn't able to communicate the issues arising from the caloric theory well. Caloric, as they used say, was not seen as something that changed form. Simply, scientist back then thought cold objects had less caloric and hot objects had more, and overall the total amount of caloric in a system does not change. This is completely against the First Law of Thermodynamics (which wasn't know at the time). Thus in Carnot's mind the heat from the cold reservoir ends up at the cold reservoir,

$$Q_{\text{Cold}} = Q_{\text{Hot}}$$

which can be thought of as the cold reservoir eventually warming up. This picture was later overturned by the development of the First Law of Thermodynamics, pioneered by scientist like Joules, which established that heat is not a conserved substance but a form of energy that can be converted into work,

$$Q_{\text{Hot}} - Q_{\text{Cold}} = W_{\text{Net}}$$

However, Carnot's theory survived almost untouched even after caloric theory was abandoned. What changed was not the mathematics, but the flawed picture of heat he so long used.

Furthermore, for his first absolute temperature scale, Kelvin imagined heat descending through temperature levels like water descending through heights, just like Carnot. So temperature was related to the "motive power" obtainable from a unit of caloric. But caloric is a false hypothesis. Thus, how the reader interprets the ratio $\frac{W}{Q}$ is totally different from what Kelvin and Carnot first realized. Kelvin thought work arose from the fall of caloric through a temperature difference, just like a glittering sunbeams reflecting off of a waterfall.

3.5.3 Kelvin's Second Absolute Temperature Scale

After becoming acquainted with James Prescott Joule's work on the mechanical equivalent of heat, Kelvin became convinced that heat and work were interchangeable manifestations of energy rather than distinct conserved quantities. This realization required him to reformulate Carnot's theory, as it was based on the caloric theory where heat was conserved as it flowed through an engine. To reformulate Carnot's theory, Kelvin considered a reversible engine operating across an infinitesimal temperature difference dT. In this limit, only an infinitesimal fraction of the transferred heat is converted into mechanical work, while the heat entering and leaving the engine differ only by an infinitesimal amount:

$$dW = Q_{\mathrm{H}} - Q_{\mathrm{C}}, \qquad Q_{\mathrm{H}} = Q_{\mathrm{C}} + O(dW)$$

To first order, the heat flowing through the engine may therefore be regarded as a single quantity,

$$Q_{\mathrm{H}} \approx Q_{\mathrm{C}} \equiv Q$$

Therefore, the heat entering and leaving the engine over an infinitesimal temperature interval are practically identical, allowing the transferred heat to be treated as a single quantity, (Q_{in}). Since the temperature difference is infinitesimal, Carnot's universal function, which generally depends on both reservoir temperatures, may be expanded as

$$\mu(T + dT, T) = \mu(T) + O(dT)$$

Because this expansion is subsequently multiplied by (dT), the higher-order terms contribute only at $(O(dT^2))$ and may be neglected. The work produced by the reversible engine can therefore be written as

$$dW = Q_{\mathrm{in}}\, \mu(T)\, dT$$

where the Carnot function now depends only on the local temperature. At this stage, however, the explicit form of ($\mu(T)$) remained unknown, and determining this function became the central challenge in Kelvin's reformulation of Carnot's theory.

Before, while Kelvin was devising his first absolute temperature scale, he had sought to figure out the Carnot Function using the temperature readings on the air thermometer. Now, however, Kelvin wondered how he would know that those numbers the air thermometer gave were the temperature. The thermometer is just assigning numbers. The air thermometer was calibrated so that it forced certain numbers, physics didn't force those numbers. What physics was telling was that an engine operating between these two states has this efficiency. Then the only thing physical is μ and we get to define temperature from that. Instead of $\mu \rightarrow \mu(T)$, Kelvin realized we can define temperature from μ such that $T = f(\mu)$. In other words, thermodynamics does not uniquely prescribe the numerical temperature scale; instead, it permits the construction of a temperature scale from the behavior of reversible heat engines. Kelvin's remaining task was therefore to determine which choice of (f) yielded the simplest and most natural formulation of thermodynamics.

Several years earlier [8], Helmholtz had proposed that Carnot's function should satisfy

$$\mu(T) \propto \frac{1}{T_{\text{abs}}}$$

where (T_{abs}) denotes the absolute temperature measured from the zero of an ideal gas thermometer (Amontons' temperature). His reasoning was motivated by the newly emerging principle that heat and mechanical work are equivalent forms of energy. If this were true, then the natural zero of temperature should correspond to a state in which no further work could be extracted from a body.

The ideal gas thermometer already provided a natural absolute scale, since the volume of an ideal gas varies linearly with temperature. A temperature measured on this scale therefore represents the distance above absolute zero. Consider two reservoirs that each cool by the same infinitesimal amount (dT). Although the temperature drop is identical in both cases, it represents a much smaller fractional change for the hotter reservoir than for the colder one:

$$\frac{dT}{T_{\text{hot}}} < \frac{dT}{T_{\text{cold}}}$$

Intuitively, lowering the temperature of a reservoir by one degree has a much greater effect when the reservoir is already close to absolute zero than when it is at a much higher temperature. Consequently, the mechanical work obtainable from a given quantity of heat should depend not on the absolute temperature drop (dT), but on the fractional temperature drop (dT/T). Since Carnot's function relates the work extracted from a quantity of heat through

$$dW = Q\,\mu(T)\,dT$$

Helmholtz was naturally led to propose that

$$\mu(T) \propto \frac{1}{T_{\text{abs}}}$$

Although this argument was heuristic rather than rigorous, it captured the essential insight that the ability of heat to produce work decreases with increasing absolute temperature. This hypothesis, later communicated to Kelvin by Joule, became the key clue in the development of the thermodynamic temperature scale.

Joule later relayed Helmholtz's proposal to Thomson. To express the conjecture in terms of the air thermometer, he wrote

$$\mu = \frac{Je}{1 + eT}$$

where (T) is the temperature measured on the Celsius scale, e is the coefficient of thermal expansion of an ideal gas ($e \approx 0.00366\ ^{\circ}\mathrm{C}^{-1}$), and J is the mechanical equivalent of heat, converting heat units into mechanical work units which is just equal to 1 since we moved to SI units. Using the relation

$$e = \frac{1}{273.15}$$

the expression may be rewritten as

$$\mu = \frac{J}{273.15 + T_{\text{C}}}$$

revealing that the denominator is simply the temperature measured from the point at which an ideal gas would extrapolate to zero volume. This temperature,

$$T_{\text{A}} = 273.15 + T_{\text{C}}$$

is known as **Amontons' absolute temperature**. It arises directly from the ideal gas relation

$$V = V_0(1 + eT_{\text{C}})$$

which predicts that the gas volume vanishes at approximately (-273.15°C). Although no real gas reaches zero volume, this extrapolated point provided the first experimentally motivated absolute temperature scale.

The significance of Joule's conjecture for Thomson was not merely the algebraic simplification. Rather, it suggested that Carnot's function might simply be inversely proportional to an absolute temperature. Since Thomson had already realized that thermodynamic temperature itself could be defined from Carnot's function, he adopted the thermodynamic temperature scale so that

$$\mu(T) = \frac{1}{T}$$

Consequently, the infinitesimal work relation became

$$\frac{dW}{Q} = \frac{dT}{T}$$

Remarkably, this thermodynamic temperature coincides with Amontons' absolute temperature (up to the choice of units), thereby unifying the thermodynamic and ideal-gas descriptions of temperature. Not deriving something else, but rather trying to operationalize the definition. This is the reason ideal gas temperature scale (using Centigrade) and Kelvin scale are translational (offset by 273.15 C degrees) [6]. It is important to remember that Kelvin could have picked any other function $f(\mu)$. For example T or $T^{34.9} - T$. The only thing that would've change would be the elegance of analytical thermodynamics of today.

There is a couple of ways a person can interpret the above results

1. $$\frac{dW}{Q} = \frac{dT}{T}$$

   For an infinitesimal reversible engine, the fraction of heat that can be converted into work equals the fractional temperature difference between the reservoirs.

2. $$dW = Q\frac{dT}{T}$$

   If the supplied heat Q is fixed, then increasing the relative temperature difference dT/T increases the work output proportionally.

3. $$\frac{dW}{dT} = \frac{Q}{T}$$

   This is not a very healthy way to look at this equation because generally heat supplied is coupled to the temperature, and you have separated the two Ts. The safest way is to see that the work output increases at a rate of Q/T joules per kelvin with respect to temperature.

Although Thomson and Joule arrived at the following relation through a different line of reasoning,

$$\frac{Q_{\text{hot}}}{|Q_{\text{cold}}|} = \frac{T_{\text{hot}}}{T_{\text{cold}}},$$

follows straightforwardly from the previous derivation for a reversible engine. Starting from the thermal efficiency,

$$\frac{W}{Q} = \frac{Q_{\text{hot}} - Q_{\text{cold}}}{Q_{\text{hot}}} = \frac{T_{\text{hot}} - T_{\text{cold}}}{T_{\text{hot}}},$$

This result has profound implications. In particular, it provides the foundation for the modern thermodynamic definition of temperature and leads naturally to the concept of entropy, which will be discussed in the next chapter. Readers interested in the historical development and practical realization of Kelvin's absolute temperature scale are encouraged to consult Chapter 5 of Chang and Yi [7].

3.5.4 Implications

This relationship is the cornerstone of Kelvin's absolute thermodynamic temperature scale and serves as the bridge from Carnot's theory of heat engines to Clausius' introduction of entropy. However, before that we must dissect this relationship. Now think about it, what does this relation tell us?

1. You should recognize is how the heat ratio is constrained by the ratio of the reservoir temperatures. The equality tells us that the amount of heat rejected is not arbitrary, and in fact it prohibits us from transforming heat entirely into work. This is the Kelvin-Planck statement:

   > "No process is possible whose sole result is the abstraction of heat from a single reservoir and the performance of an equivalent amount of work." [6]

   You might object and suggest what if T_{cold} is 0K, that would allow W/Q = 1 right ? The two famous statements of the second law by Clausius and Kelvin, often introduced in thermodynamics textbooks, were originally postulates rather than proven results. During their time, neither Clausius nor Kelvin had a deeper theoretical framework from which these statements could be derived. Instead, they proposed them as fundamental principles based on the observed behavior of heat transfer and the limitations of heat engines. A reader with knowledge of Heisenberg's Uncertainty principle might think: "If we drained all the thermal energy, wouldn't a particle become perfectly stationary, giving it a precise position and velocity, which quantum mechanics forbids?" Indeed, but even at T=0K, a quantum particle still possesses ”zero-point energy.” It is still ”moving” in a quantum sense. You might object again: ”But if there is movement, there must be temperature! Isn't temperature just the average movement of particles?” Classically, yes. But thermodynamics reaches deeper. It defines temperature not by speed, but by how entropy (disorder) changes with internal energy:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}$$

Temperature measures how eager a system is to share its energy. Nothing in this definition says particles must stop moving. Textbooks don't always spell this out, but the absolute ban on reaching T=0K is so fundamental that we call it the Third Law of Thermodynamics (specifically, the Unattainability Principle). It states that no physical process can ever cool a system to absolute zero in a finite number of steps.

2. Secondly, efficiency of a reversible engine depends only on the temperature of its reservoirs the engine is operating between.

$$\eta = 1 - \frac{T_{\mathrm{C}}}{T_{\mathrm{H}}}$$

3. Just as Carnot observed, higher temperature heat has a higher work producing potential.

$$W = Q_{\mathrm{hot}} - Q_{\mathrm{cold}} = Q_{\mathrm{hot}}\left(1 - \frac{T_{\mathrm{cold}}}{T_{\mathrm{hot}}}\right)$$

Even if the reservoir has 1kJ, when it is supplied at a higher temperature, the efficiency increases. This is very strange. And the reason you are thinking this way is because you are still stuck with the accounting mathematics of the First Law, while we are in the presence of the Second Law.

There are certainly many more implications and benefits (such as Clasius-Clapeyron equation and alternate derivation for Stefan's Law) of this relationship, but these are the ones I wished to highlight here.

Before moving on, I would like to offer a brief teaser for the next chapter on Clausius. In the previous section, we introduced the concept of a state function. We have now shown that, for a reversible cycle,

$$\frac{Q_{\mathrm{hot}}}{T_2} + \frac{Q_{\mathrm{cold}}}{T_1} = 0.$$

More generally, this may be written as

$$\oint_{\mathrm{rev}} \frac{\delta Q}{T} = 0.$$

The reader may recognize this as Clausius' celebrated result, from which the existence of a new thermodynamic state function naturally follows. Clausius named this state function entropy, and in the next section we will see how this seemingly simple equation fundamentally changed the way physicists understood heat, work, and the second law of thermodynamics.

3.6 Clausius Insists On Rum

Rudolf Clausius's is one of the pillars of thermodynamics. His paper, On the Moving Force of Heat and the Laws of Heat which may be Deduced Therefrom, establishes the mechanical theory of heat equivalent of Carnot's principle, firmly states the first law of thermodynamics, and introduces the second law of thermodynamics.

In this section, we will first see how Clausius naturally derived Kelvin's $1/T$ conclusion, then we will move to the more pressing matter of how he realized the second law of thermodynamics.

3.6.1 $1/T$ and Internal Energy By Clausius

For an arbitrary fluid let's assume a reversible engine, an infinitesimal Carnot cycle [2].

Isothermal Expansion ($a \to b$)

First we operate over an isotherm where the volume changes infinitesimally and heat is injected into our system from the constant temperature reservoir ($dT = 0$).

$$dQ_{ab} = \lambda(T, V)\, dV$$

where λ is the latent heat.

Adiabatic Expansion ($b \to c$)

During the adiabatic expansion latent heat energy is transformed to work without any heat exchange, thus $dQ_{bc} = 0$. So the temperature decreases from T to $T - dT$, and the volume changes by an amount $\delta' V$. Therefore,

$$0 = -C_V(T, V + dV)\, dT + \lambda(T, V + dV)\, \delta' V.$$

To first order,

$$\delta' V = \frac{C_V(T, V + dV)}{\lambda(T, V + dV)}\, dT.$$

Isothermal Compression ($c \to d$)

The system is then compressed isothermally at the lower temperature $T - dT$. The heat rejected is

$$dQ_{cd} = -\lambda(T - dT, V + \delta V)\, d'V,$$

where the width of the lower isotherm is

$$d'V = dV + \delta' V - \delta V.$$

Expanding $\lambda(T - dT, V + \delta V)$ about (T, V) and retaining terms up to order $dTdV$,

$$dQ_{cd} = -\lambda d'V + \frac{\partial \lambda}{\partial T} dTdV.$$

Substituting the expression for $d'V$ gives

$$dQ_{cd} = -\lambda dV - \lambda(\delta' V - \delta V) + \frac{\partial \lambda}{\partial T} dTdV.$$

Adiabatic Compression ($d \to a$)

Finally, the system is compressed adiabatically back to its initial state, so that $dQ_{da} = 0$. The temperature increases from $T - dT$ to T, while the volume changes by an amount δV. Hence,

$$0 = C_V(T, V)\, dT - \lambda(T, V)\, \delta V,$$

or

$$\delta V = \frac{C_V(T, V)}{\lambda(T, V)}\, dT.$$

Similarly,

$$\delta' V = \frac{C_V(T, V + dV)}{\lambda(T, V + dV)}\, dT.$$

Expanding the latter about (T, V) yields

$$\delta' V - \delta V = \frac{\partial}{\partial V} \left(\frac{C_V}{\lambda} \right) dTdV.$$

Substituting this result into the expression for dQ_{cd} gives

$$dQ_{cd} = -\lambda dV + \left(\frac{\partial \lambda}{\partial T} - \frac{\partial C_V}{\partial V} \right) dTdV.$$

Internal Energy

Adding the heat exchanged during the two isothermal processes,

$$dQ_{ab} + dQ_{cd} = \lambda dV + \left[-\lambda dV + \left(\frac{\partial \lambda}{\partial T} - \frac{\partial C_V}{\partial V}\right) dTdV\right]$$
$$= \left(\frac{\partial \lambda}{\partial T} - \frac{\partial C_V}{\partial V}\right) dTdV$$

The heat absorbed from the hot reservoir during the isothermal expansion is λdV , while the net heat exchanged over one complete cycle is given by the expression above. By the First Law of Thermodynamics, the net heat exchanged must equal the net work performed $dQ_{\text{cycle}} = dW_{\text{cycle}}$, where work performed by the engine is equal to the area enclosed by the infinitesimal Carnot cycle $dW_{\text{cycle}} = dPdV$. We can re-write the work differential as

$$dPdV = \left(\frac{\partial P}{\partial T}\right)_V dTdV$$

$$\left(\frac{\partial \lambda}{\partial T} - \frac{\partial C_V}{\partial V}\right) dTdV = \left(\frac{\partial P}{\partial T}\right)_V dVdT$$

Equivalently, $$\frac{\partial(\lambda - P)}{\partial T} - \frac{\partial C_V}{\partial V} = 0$$

which is the integrability condition for the differential form

It is worth pulling one more result out of this derivation before moving on, because it is easy to walk past it. The integrability condition itself,

$$\frac{\partial(\lambda - P)}{\partial T} = \frac{\partial C_V}{\partial V},$$

combined with $C_V = (\partial U/\partial T)_V$ and $U = U(T, V)$, forces $\lambda = T(\partial P/\partial T)_V$. This relation, sometimes called the thermodynamic equation of state, looks like a piece of bookkeeping, but it is not. It is a Maxwell relation in disguise, decades before Maxwell or the potentials that bear his relations were formulated. Once entropy is properly defined in the next section, this same identity reappears as $(\partial S/\partial V)_T = (\partial P/\partial T)_V$, one of the four Maxwell relations we will meet again when we build the thermodynamic potentials. What Clausius actually found here, without yet having the word for it, was the second law expressing itself as a constraint between two directly measurable quantities, λ and $(\partial P/\partial T)_V$. Keep this result in your pocket. It will surface again.

Moving on, the line integral of this differential depends only on the initial and final equilibrium states. In particular, the integral around any closed path must vanish,

$$\oint \left[C_V\, dT + (\lambda - P)\, dV\right] = 0.$$

showing that the mixed partial derivatives are equal. This is analogous to a curl-free vector field in vector calculus: the integral of the differential depends only on the initial and final equilibrium states, not on the path taken between them. In other words, the differential is exact, implying the existence of a state function. Clausius therefore identified this exact differential with a new thermodynamic state function, the internal energy,

$$dU = C_V dT + (\lambda - P)dV = dQ - PdV$$

Such that,

$$U(T,V) - U(T_0,V_0) = \int_{(T_0,V_0)}^{(T,V)} \left[C_V\, dT + (\lambda - P)\, dV\right] = \int_{(T_0,V_0)}^{(T,V)} (dQ - P\, dV)\,.$$

The introduction of the internal energy transformed the First Law from a statement about cyclic heat engines into a general relation applicable to any thermodynamic process.

Efficiency

Having established the existence of the internal energy, Clausius next turned his attention to the efficiency of a reversible Carnot engine. To simplify the analysis, he considered an ideal gas. For an ideal gas, he assumed correctly that the internal energy depends only on the temperature,

$$U = U(T).$$

Consequently,

$$dU = C_V\, dT.$$

Comparing this result with the previously derived expression,

$$dU = C_V\, dT + (\lambda - P)\, dV,$$

immediately yields $\lambda = P$. This result has a simple physical interpretation. During an isothermal expansion of an ideal gas, the supplied heat does not alter the microscopic energy of the gas. Instead, it is converted entirely into expansion work. Therefore, the heat absorbed during the upper isotherm is simply

$$Q_H = \lambda\, dV = P\, dV.$$

The efficiency of the infinitesimal Carnot cycle is defined as the ratio of the work performed to the heat absorbed,

$$\eta = \frac{W}{Q_H}.$$

Since the work performed by the infinitesimal cycle is equal to the area enclosed by the cycle,

$$W = dP\,dV,$$

the efficiency becomes

$$\eta = \frac{dP\,dV}{P\,dV} = \frac{dP}{P}.$$

For an ideal gas,

$$PV = R(T + 273.15),$$

where T denotes the Celsius temperature used by Clausius. Holding the volume constant,

$$dP = \frac{R}{V}\,dT.$$

Dividing by the ideal-gas equation gives

$$\frac{dP}{P} = \frac{dT}{T + 273.15},$$

Thus,

$$\eta = \frac{dT}{T + 273.15}.$$

Carnot's theorem states that the efficiency of every reversible engine operating between two infinitesimally separated temperatures depends only on those temperatures and not on the working substance. Clausius therefore introduced a universal function $F(T)$ such that

$$\eta = F(T)\,dT.$$

Comparing this expression with the result obtained above immediately gives

$$F(T) = \frac{1}{T + 273.15}.$$

This universal function as we know is the Carnot function. Its appearance marks an important conceptual advance: the efficiency of every reversible heat engine is governed by a universal law independent of the material composing the engine. By redefining the temperature scale so that

$$T_{\text{abs}} = T + 273.15,$$

the Carnot function assumes the remarkably simple form

$$F(T_{\text{abs}}) = \frac{1}{T_{\text{abs}}}.$$

It is worth stopping here and noticing what just happened. This is the third time in this chapter that the same relation has fallen out of an argument, and each time it fell out of a completely different kind of reasoning. Kelvin found $\mu(T) = 1/T$ by asking what definition of temperature makes Carnot's function simplest. Joule and Thomson found $Q_{\text{hot}}/Q_{\text{cold}} = T_{\text{hot}}/T_{\text{cold}}$ from calorimetric measurement and Helmholtz's conjecture. Clausius has now found $F(T_{\text{abs}}) = 1/T_{\text{abs}}$ from nothing but an infinitesimal Carnot cycle for an arbitrary fluid and the integrability of an exact differential. Three independent routes, three different starting assumptions, one destination. When that happens in physics it is rarely a coincidence, and it is usually a sign that you have found something more fundamental than any one of the arguments that led you there. In fact this quantity later becomes the integrating factor that Clausius defines as entropy

$$dS = \frac{dQ_{\text{rev}}}{T_{\text{abs}}}.$$

3.6.2 Deriving $Q_h/T_h = Q_c/T_c$

Everything derived so far, $dW/Q = dT/T$, was proven only for an infinitesimal Carnot engine, one operating between two reservoirs an infinitesimal distance dT apart. A real Carnot engine operates between two reservoirs at finite temperatures T_h and T_c, possibly hundreds of degrees apart, and nothing said so far tells us how the finite engine behaves. Filling this gap is not an extra assumption. It is exactly what it means for the finite engine to be reversible.

Recall the very first section of this paper: a reversible process is a succession of equilibrium states, each infinitesimally close to the last. A finite Carnot cycle, if it is genuinely reversible, is therefore nothing more than a continuous chain of infinitesimal Carnot cycles, stacked one after another, spanning the full range from T_h down to T_c. Picture this literally. The first infinitesimal engine in the chain absorbs heat Q_h from the hot reservoir at T_h, does infinitesimal work, and rejects its leftover heat, not to some external reservoir, but directly into the next infinitesimal engine in the chain, which now sits at temperature $T_h - dT$. Because this handoff happens at matched, infinitesimally close temperatures, no finite temperature gap ever opens up between stages, and no irreversibility is smuggled in at the joints. The chain continues, stage by stage, until the last infinitesimal engine rejects its remaining heat Q_c to the cold reservoir. The heat flowing through this chain is a continuous function of the local temperature, $Q(T)$.

Now apply the relation already derived for a single infinitesimal stage. As the heat passes through the stage sitting at temperature T, an infinitesimal amount of work dW is extracted, and by energy conservation the heat leaving that stage is smaller than the heat entering it by exactly that amount. Writing dQ for the (positive) decrease in Q as the chain descends through a temperature drop dT,

$$dQ = dW,$$

and substituting the infinitesimal relation $dW = Q(T)\,\dfrac{dT}{T}$ derived earlier gives a single, separable differential equation governing how the heat in the chain falls off with temperature,

$$\frac{dQ}{Q} = \frac{dT}{T}.$$

This equation says something simple: at every stage of the chain, the fractional loss of heat equals the fractional drop in temperature. Integrating both sides from the top of the chain to the bottom,

$$\int_{Q_h}^{Q_c} \frac{dQ}{Q} = \int_{T_h}^{T_c} \frac{dT}{T} \quad \Longrightarrow \quad \ln\left(\frac{Q_c}{Q_h}\right) = \ln\left(\frac{T_c}{T_h}\right),$$

and therefore

$$\frac{Q_c}{Q_h} = \frac{T_c}{T_h}$$

This is the result we were after, and notice what made it possible. We never analyzed a finite-temperature-difference engine directly, nor did we need to. We only ever used the infinitesimal relation, applied at every point along a continuous chain, and let integration carry that infinitesimal statement all the way from T_h to T_c. This is the same idea that will reappear, in a more general form, when we extend this result beyond a simple two-reservoir engine to an arbitrary reversible cycle exchanging heat with a continuum of temperatures at once, rather than just two.

3.6.3 From a Special Case to a Universal Law: The Clausius Inequality

At the end of the previous section we found

$$\frac{Q_c}{T_c} + \frac{Q_h}{T_h} = 0 \ \Rightarrow \ \oint_{\text{rev}} \frac{\delta Q}{T} = 0,$$

but only for a reversible cycle. What happens when the cycle is irreversible?

Step 1: Replace every reservoir with one helper, connected to a single T_0

A real cycle touches many different temperatures as it runs, hot ones when it absorbs heat, cold ones when it rejects heat. Clausius's move was to stop dealing with all of them directly. Instead, pick one fixed reservoir at temperature T_0, and for every point where the real cycle touches some local temperature T, insert a small reversible Carnot engine between T_0 and T to handle that exchange.

Here is the part that usually causes confusion, so it is worth stating plainly: this auxiliary Carnot engine does different jobs depending on what the real cycle is doing at that moment.

- When the real cycle is absorbing heat at some T, the helper engine acts like an ordinary engine: it draws heat from T_0, sends it down to T, and produces work.
- When the real cycle is rejecting heat at some T, the same helper engine runs the other way, as a refrigerator: it takes the rejected heat from T and pumps it back up to T_0, consuming work to do it.

So T_0 is not only a source. It is also where all the rejected heat ends up, just indirectly, carried back by the helper engine instead of being dumped into some separate third reservoir. You do not need a second external reservoir for the rejected heat to go to, because the helper engine's whole job is to shuttle heat back and forth between the real cycle and T_0 in both directions, as needed.

By Carnot's theorem, every reversible engine between two fixed temperatures has the same efficiency, so the ratio of heat exchanged is fixed,

$$\frac{Q_0}{Q} = \frac{T_0}{T},$$

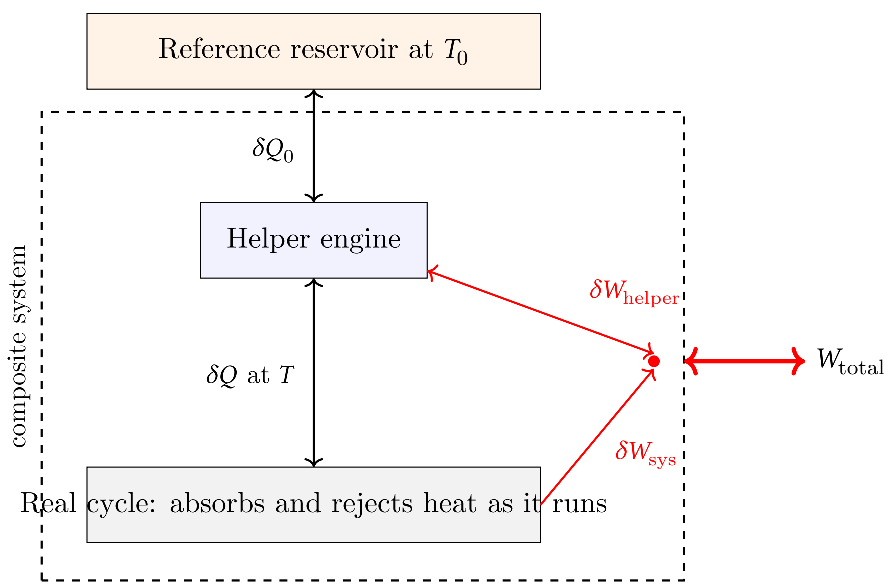
no matter the engine's size. The only free choice is how much heat to push through it. So for each small exchange δQ the real cycle needs at temperature T, size a helper engine to match

$$\delta Q_0 = T_0 \frac{\delta Q}{T}.$$

This formula already has the right sign built in. If $\delta Q > 0$ (absorbing), then $\delta Q_0 > 0$ too, heat really is flowing from T_0 down to the system. If $\delta Q < 0$ (rejecting), $\delta Q_0 < 0$ as well, heat is flowing from the system back up to T_0. One formula, both directions, automatically.

Step 2: Draw a box around everything except T_0

Now treat the real cycle plus every one of these helper engines as one machine.



The double-headed arrows are deliberate, every quantity in this picture can point either way, depending on where the real cycle is in its loop. What stays fixed is the boundary. One instance of this picture sits at every point around the real cycle, and every single one of them is drawn inside the dashed box. The exchange between a helper engine and the real system, at whatever local T they happen to be touching, never leaves the box. It is purely internal.

The only thing crossing the dashed boundary, at any point, is heat to or from T_0, and work. Add up every helper's contribution over one full lap of the cycle, and the total heat exchanged with T_0 is

$$T_0 \oint \frac{\delta Q}{T}.$$

Step 3: That heat becomes work, exactly

Each helper engine is cyclic, so its own first law reads

$$\delta Q_0 = \delta W_{\text{helper}} + \delta Q \implies \delta W_{\text{helper}} = \delta Q_0 - \delta Q.$$

The real system is also cyclic, so over one full lap,

$$W_{\text{sys}} = \oint \delta Q.$$

The heat leaving a helper engine and the heat the system receives at that same moment are the same transfer, counted twice, once as an outflow and once as an inflow. Add up all the work in the composite system:

$$W_{\text{total}} = \oint \delta W_{\text{helper}} + W_{\text{sys}} = \oint (\delta Q_0 - \delta Q) + \oint \delta Q = \oint \delta Q_0 = T_0 \oint \frac{\delta Q}{T}.$$

The internal δQ cancels exactly, every time, because it was never a separate quantity to begin with. What's left is only the heat that actually crossed the outer boundary, from T_0.

Step 4: Kelvin's statement forces the inequality

A machine that takes heat from one reservoir and turns all of it into work, nothing else happening, is exactly what Kelvin's statement forbids, a statement we already showed is equivalent to Clausius's own. Since $T_0 > 0$, ruling out $W_{\text{total}} > 0$, due to contradiction otherwise, forces

$$\boxed{\oint \frac{\delta Q}{T} \leq 0}$$

Step 5: Reversible cycles hit the bound exactly

If the real cycle is reversible, run the whole thing backward, real cycle and every helper engine together, since all of them are reversible. This flips every heat flow, $\delta Q \to -\delta Q$, so the same argument on the reversed cycle gives

$$\oint \frac{-\delta Q}{T} \leq 0 \implies \oint \frac{\delta Q}{T} \geq 0.$$

Together with Step 4, a reversible cycle satisfies both ≤ 0 and ≥ 0 at once, which forces

$$\oint_{\text{rev}} \frac{\delta Q}{T} = 0,$$

the earlier result. For a real, irreversible cycle, the inequality is strict

$$\oint \frac{\delta Q}{T} < 0.$$

This is the Clausius inequality [63].

3.6.4 Defining Entropy for Irreversible Processes

The Clausius inequality is what finally lets entropy escape the narrow confines of reversible cycles. Take any process that carries a system irreversibly from state 1 to state 2, and imagine returning it to state 1 by any convenient reversible path. Together these two legs form a complete cycle, so the Clausius inequality applies to the whole loop,

$$\int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{irrev}} + \int_2^1 \left(\frac{\delta Q}{T}\right)_{\text{rev}} \leq 0.$$

The second integral is over the reversible return leg, from state 2 back to state 1, and by the definition established in the previous section this is exactly the entropy of the starting point of that leg minus the entropy of its endpoint,

$$\int_2^1 \left(\frac{\delta Q}{T}\right)_{\text{rev}} = S(1) - S(2).$$

Substituting this in,

$$\int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{irrev}} + \left[S(1) - S(2)\right] \leq 0.$$

Now isolate the irreversible integral by adding $S(2) - S(1)$ to both sides. This step is worth doing carefully, because it is easy to lose track of a sign here. Adding the same quantity to both sides of an inequality never flips it, unlike multiplying by a negative number, so the direction of $\leq$ is preserved exactly as written:

$$\int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{irrev}} \leq S(2) - S(1).$$

Writing $\Delta S \equiv S(2) - S(1)$ for the entropy change of the actual irreversible process (entropy is a state function, so this is well-defined regardless of how the system actually got from 1 to 2).

$$\boxed{\Delta S \geq \int_1^2 \frac{\delta Q}{T}}$$

This is the entire content of the result: the inequality did not change direction anywhere in the algebra, we only moved a term from one side to the other and then relabeled which side we chose to write first.

With equality recovered only in the reversible limit, this single inequality is, in a real sense, the entire second law compressed into one line: for any real, irreversible process, entropy increases by <u>more</u> than the heat exchanged divided by temperature would naively suggest. The gap between the two sides is not a bookkeeping error, it is physically produced entropy, and for a thermally isolated system ($\delta Q = 0$) the inequality collapses to the version every student eventually memorizes without necessarily seeing where it came from,

$$\Delta S_{\text{isolated}} \geq 0.$$

It is worth pausing on how little was assumed to get here. We did not assume anything new about molecules, disorder, or probability, nothing Clausius himself had any access to. Everything above follows from Kelvin's statement of the second law and the existence of the state function established two sections ago. The statistical picture, where entropy counts microstates, is still a full chapter away in this paper, and Clausius reached this result without it.

Having established that entropy exists as a state function and that it can only increase for any process an isolated system actually undergoes, Clausius distilled the first and second laws together into a single, famous sentence, closing his 1865 paper with what remains the most quoted line in the history of thermodynamics:

> Die Energie der Welt ist constant. Die Entropie der Welt strebt einem Maximum zu.
> "The energy of the universe is constant. The entropy of the universe tends to a maximum." [62]

Read carelessly, this sounds like a throwaway flourish, a bit of cosmic poetry appended to an otherwise technical paper. It is not. It is a precise claim, and it follows directly from everything derived above: treat the universe as a single isolated system (nothing lies outside it to exchange heat or work with), apply the first law to it (its total energy cannot change) and apply the entropy-generation inequality to it (its entropy cannot decrease), and the sentence above is simply the two results stated in one breath. It is also the sentence that, read a certain way, first raised the question of the "heat death" of the universe, the idea that if entropy really does climb toward a maximum without limit, the universe approaches a final state in which no further work can be extracted from anything, anywhere, ever again, a conclusion Clausius himself explored and one that unsettled a great many of his contemporaries far more than the mathematics that produced it.

3.6.5 $\tau\rho o\pi\acute{\eta}$ for Transformation

Having proven that this new quantity exists and behaves lawfully, Clausius needed a name for it, and the name he chose is worth dwelling on because it tells you what he thought he had found. In his 1865 paper, he wrote that he deliberately modeled the new word on "energy" itself, choosing the Greek root $\tau\rho o\pi\acute{\eta}$ (tropē, meaning transformation) precisely because he wanted a word that would sit beside Energie in the vocabulary of physics and signal a comparably fundamental status [62]. This was not incidental branding. Clausius had spent the previous decade developing what he called the "equivalence of transformations," the idea that every thermodynamic process involves two distinct bookkeeping quantities: how much energy is conserved (the first law), and how much of that energy's ability to do further work has been irreversibly transformed, or as he put it, degraded. Entropy was his name for a precise, quantitative measure of the second kind of change. Energy tells you how much you have. Entropy, in Clausius's own conception, tells you how much of what you have is still capable of producing motive power.

Entropy did not appear from nowhere in 1865. Three years earlier, Clausius had already been groping toward the same idea under a cruder name: disgregation [63][64]. His idea, developed while thinking about interior work done during changes of state such as melting or vaporization, was that the arrangement of the constituent parts of a body, how spread out or how tightly bound its particles are, constitutes a distinct physical quantity in its own right, separate from the ordinary mechanical work done against external pressure. Melting ice, for instance, absorbs latent heat while doing essentially no external work, and Clausius wanted a bookkeeping quantity that would account for where that heat had gone: not into moving a piston, but into loosening the internal arrangement of the molecules against their own mutual attraction. It was a qualitative, almost intuitive placeholder, defined in terms of a body's internal configuration rather than derived from any general principle, and Clausius never fully succeeded in making it as rigorous or as universally applicable as entropy eventually became [64][66]. But the instinct behind it, that there ought to exist a quantity measuring how dispersed or disordered the internal arrangement of a system is, turned out to be exactly right. It is not a coincidence that this is the same intuition Boltzmann would later make precise, quantitative, and universal through his statistical definition of entropy in terms of the number of accessible microstates. The connection between entropy and disorder was not a later reinterpretation grafted onto Clausius's purely macroscopic, engine-based derivation; it was there in his own thinking from the very beginning. He simply lacked the statistical machinery to make it precise.

A handful of Clausius's other contributions round out the picture of how much of the standard machinery of thermodynamics traces back to this one person. He gave the second law its cleanest verbal form, the statement we proved equivalent to Kelvin's earlier in this chapter: heat does not spontaneously flow from a colder body to a hotter one [63]. He was also the first to state the first law of thermodynamics with full generality as a statement about internal energy as a state function, rather than as a special property of cyclic heat engines, exactly the result we derived from his infinitesimal Carnot cycle argument two sections ago. His name is also permanently attached to the Clausius–Clapeyron equation, governing the slope of a phase boundary on a P–T diagram, already derived and discussed in the chapter on thermodynamic potentials.

What ties all of this together, and what makes Clausius arguably the single most consequential figure in this chapter, is that he took Carnot's engine-room reasoning, stripped of its dependence on the false caloric theory, and turned it into a general, exact mathematical framework: internal energy as a state function, entropy as a state function, and a single inequality governing how the second one behaves, applicable to absolutely any thermodynamic system, not merely to idealized heat engines. Kelvin gave temperature its universal, substance-independent meaning. Clausius gave the second law its universal, process-independent meaning. But notice what Clausius's framework can and cannot do. It can tell you, with total mathematical certainty, that entropy increases. It cannot tell you why nature insists on this, what entropy actually is at the level of atoms and molecules, or why disgregation, Clausius's own half-formed intuition about arrangement and dispersal, happened to be pointing in the right direction all along. Clausius built the law. He never saw the mechanism underneath it. That mechanism, and the statistical picture that finally explains why the second law holds at all, is where Boltzmann takes this story next.

4 Enthalpy

4.1 Introduction

Why did I decided to write about enthalpy and why should you read? This is a question you must be wondering especially since it seems so trivial doesn't it. Well there are few things that took me a while to understand about enthalpy in thermodynamics. There are loads of things professors say that are practically correct but technically incomplete. In engineering this is the case as engineers deal with the practical. However, for a more curious mind, certain technicalities might get in the way. For example imagine a system which can be absolutely anything, a wooden block, trapped gas, the vodka you carry in your bottle to your next class. Now we are told that the internal energy of an ideal gas depends only on its temperature. A natural question then arises: what happened to the enormous pressure exerted by the surrounding atmosphere? That pressure continuously acts on the system and stores mechanical energy in its deformation. If mechanical work can ultimately be converted into thermal energy, why does the internal energy appear to ignore it? Furthermore, if the pressure of the surroundings already contributes to the system's energy, why do we introduce a new thermodynamic quantity,

$$H = U + PV,$$

instead of simply incorporating everything into the internal energy?

These are precisely the questions that motivated this chapter. Rather than treating enthalpy as a convenient mathematical definition, we will examine why it is introduced, what physical problem it solves, and why it naturally appears whenever processes occur at constant pressure.

Lets first build why and where Enthalpy is useful.

4.1.1 Why Enthalpy Appears At All

Enthalpy is the energy associated with the work done in moving the system's boundary, by the surrounding environment, in addition to the internal energy already stored in the system [67]. This definition, though, invites a fair question: how much does that PV term actually matter in practice? We know the pressure, usually just atmospheric, but we also routinely treat solids and liquids as incompressible, and even for a gas, if the system does not change elevation, PV often looks unchanged between inlet and outlet. So why bother carrying it at all?

For a closed system containing an incompressible substance, a liquid or a solid, the answer is often. It genuinely doesn't matter much. Starting from the definition,

$$\begin{aligned} H &= U + PV \\ dH &= dU + V\,dP + P\,dV. \end{aligned}$$

Since the substance is incompressible, $dV = 0$, and at constant pressure $dP = 0$ as well, so $dH = dU$ exactly. In this setting, the distinction between H and U really is just bookkeeping: recall from the first law that it is dU or dH that governs energy balances (not H or U), and for a closed system at constant pressure the two track each other so closely that the choice between them is largely academic. Moreover, even for heating a compressible substance in closed system,

$$\begin{aligned} dH &= dU + V\,dP + P\,dV \\ dU &= \delta Q - PdV \\ \text{Thus,} \qquad dH &= \delta Q - VdP \end{aligned}$$

and if the pressure is constant, $dH = \delta Q$.

But this is precisely where the skepticism above stops being justified, because a closed system is a special case, not the general one. The moment mass itself crosses the system boundary, rather than only heat and work, the picture changes completely, and the PV term stops being bookkeeping and becomes physically indispensable.

Picture a pipe with fluid flowing steadily through some device, a pump, a turbine, a section of pipe, anything. Focus on the fluid that is about to enter the device. Right behind it sits more fluid, and that fluid has to shove the parcel in front of it across the boundary to make room for itself. Pushing a volume V_{in} across a boundary against a pressure P_{in} costs work,

$$W_{\text{flow,in}} = P_{\text{in}} V_{\text{in}}.$$

This is not the same kind of work as the boundary work $\int P\,dV$ from a piston expanding, and it is worth being careful about the distinction, because it is easy to blur them together. Boundary work happens because the system's own volume is changing. Flow work happens because mass is crossing a fixed boundary regardless whether or not the system's own volume changes. A steady flow through a rigid pipe has zero boundary work everywhere, the control volume's shape never changes, and yet flow work is happening continuously at both the inlet and the outlet, because fluid keeps being shoved in and shoved out.

The same logic applies at the exit. To leave the control volume, the fluid parcel must push the fluid already downstream out of the way, doing work $P_{\text{out}} V_{\text{out}}$ on its surroundings. So every unit of mass that crosses the boundary carries two things with it, not one: its own internal energy U, and the flow work PV that was either spent pushing it in or that it spent pushing its way out.

This is the reason enthalpy is not bookkeeping for an open system the way it can look like bookkeeping for a closed one. Write the steady-flow energy balance for a control volume, tracking the energy that flows in and out with the mass itself, and the internal energy and the flow work do not appear as two separate terms you could choose to ignore. They appear glued together, because that is literally what a flowing parcel of fluid is carrying across the boundary, per unit mass,

$$u + Pv \equiv h,$$

where v is specific volume. This combination is not something we chose for convenience, it is forced on us the moment mass, rather than only energy, crosses the system boundary [67][22]. The full steady-flow energy balance for a single inlet, single outlet control volume is

$$\dot{Q} - \dot{W}_{\text{shaft}} = \dot{m}\left[(h_{\text{out}} - h_{\text{in}}) + \frac{V_{\text{out}}^2 - V_{\text{in}}^2}{2} + g(z_{\text{out}} - z_{\text{in}})\right],$$

and it is worth noticing exactly what does and does not appear here. Elevation enters through its own separate term, $g\Delta z$, entirely unrelated to Pv. If a system does not change elevation, the potential energy term vanishes, correctly, but this has nothing to do with whether the Pv term matters. Flow work is about pressure and specific volume at the boundary, not about height, and conflating the two is an easy trap to fall. For example, if the entire energy driving a liquid down a pipe is ultimately supplied by gravity, the (Pv) term should not be regarded as a separate source of energy. Rather, it is part of the bookkeeping required to account for the energy carried by the fluid as it crosses the control surface, thus Pv only exists at the boundaries where the fluid crosses the control surface.

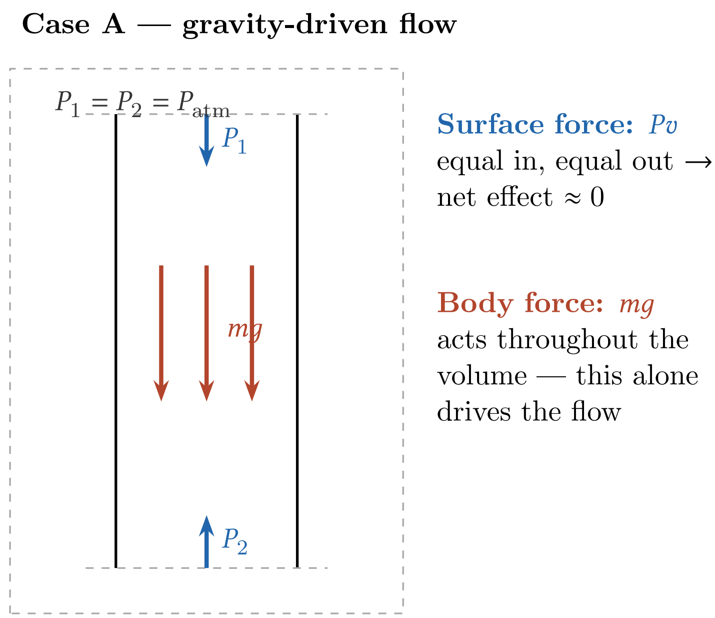


Figure 1: A control volume in a vertical pipe with equal pressure at the inlet and outlet. The flow-work terms at the boundary cancel, but gravity, acting throughout the volume, still drives the fluid downward.

This also finally answers the "why do we care" question for an incompressible liquid or solid. Take a pump moving an incompressible liquid from a low pressure P_1 to a higher pressure P_2, with negligible heat transfer and negligible change in internal energy, since the liquid is neither heating up nor changing phase. If you tried to account for this using U alone, you would conclude, correctly, that $\Delta U \approx 0$, and wrongly, that therefore almost no energy was needed to run the pump. But the shaft work actually required by a real pump is not zero, and it is exactly

$$w_{\text{pump}} = \Delta h = v\,(P_2 - P_1),$$

since v is constant for an incompressible fluid and $\Delta u \approx 0$. Every bit of that pump work went into flow work, shoving the same volume of liquid from a region of low pressure into a region of high pressure, and ΔU was structurally incapable of seeing it, because ΔU never had access to information about the pressure at the boundary in the first place. This is the honest answer to the skepticism this section opened with. The PV term is not something we carry around out of habit or convenience. For a closed system with no mass crossing the boundary, it genuinely often is negligible bookkeeping, exactly as argued above. For a flow process, it is the entire physical mechanism by which a pump, compressor, or turbine does its job, and dropping it does not simplify the accounting, it deletes the real work term from the equation.

4.1.2 Where did the atmosphere's pressure go?

Lets get back to some of our questions now, starting with the first question. It is tempting to imagine the atmosphere as constantly loading the system with mechanical energy, the way a spring stores energy the moment you compress it, and to worry that U is somehow blind to this. But notice what work actually requires: a force acting through a displacement. A static pressure sitting on a system that is not changing volume is doing precisely zero work on it, full stop, no matter how large that pressure is. One atmosphere pressing on a steel block is enormous in absolute terms, roughly 10^5 N/m^2, but if the block's volume never budges, that pressure never moves anything, and $F\,dx = 0$ for every $dx = 0$. There is no mechanical energy being continuously stored, because storing energy this way requires ongoing deformation, and ongoing deformation is exactly what an incompressible substance, by definition, refuses to do.

Where the atmosphere's pressure does do something is in setting the equilibrium state the system actually sits in. A real solid or liquid is never perfectly incompressible, it strains by some tiny amount under one atmosphere, and that strain does store a small amount of elastic energy. But this energy is not missing from U, it is already inside U, because U is defined as a function of the system's actual state, and the actual state already reflects whatever tiny compression the surrounding pressure produced. Internal energy was never blind to pressure; it changes because pressure changes the equilibrium volume, and with it the potential energy stored between molecules, and because pressure changes the equilibrium temperature, and with it the heat content of the substance. For a genuinely incompressible substance, that indirect dependence vanishes because the volume literally cannot respond, and for a real, very-nearly-incompressible substance like water or steel under ordinary atmospheric loading, the resulting elastic energy is real but numerically negligible, which is precisely why treating ΔU as pressure-independent for solids and liquids is a good approximation rather than an oversight. For example, a 1 m^3 block of pure iron taken from Earth into space at constant temperature would expand by only 0.0000596% of its original volume.

An ideal gas makes the same point from the opposite direction and is worth pausing on, because it looks at first like a counterexample. An ideal gas is highly compressible, and yet its internal energy depends on temperature alone, $U = U(T)$, with no dependence on P or V whatsoever. This is not U ignoring the atmosphere either. It follows from the model itself: an ideal gas has no intermolecular attraction or repulsion, so there is no potential energy stored between molecules for pressure to modify, only kinetic energy, which is set by temperature through equipartition (see Maxwell's Kinetic Theory). The pressure a gas exerts on a piston is a real, physical bombardment of molecules transferring momentum, and it is entirely responsible for PdV work when the piston actually moves, but it does not correspond to any hidden reservoir of stored energy sitting inside the gas waiting to be accounted for. The energy transfer only happens, and only needs bookkeeping, at the moment the boundary surface actually moves.

4.1.3 Why introduce H instead of folding everything into U

Suppose you burn something in a beaker open to the air, or run a reaction in a flask, or let steam expand through a turbine. In every one of these cases you want to know one thing: how much heat was released or absorbed. You measure Q. But Q is not ΔU, because as the system changes it also pushes against the atmosphere (or does work on the turbine blades), so some of the energy leaving the system left as boundary work, not as heat. To recover ΔU from a measured Q, or to predict Q from a computed ΔU, you have to separately track $P\Delta V$ and subtract it off, by hand, every single time.

This would be a minor nuisance if constant-pressure processes were rare but they are not. A sealed, rigid bomb calorimeter is the exception in the laboratory; an open beaker, an open flask, a furnace venting to atmosphere, a turbine sitting between two roughly fixed pressures, is the rule everywhere else, in chemistry, in engineering, in biology. So this correction is not an occasional annoyance, it is a tax paid on nearly every real measurement and every real process, forever, unless something is done about it.

The fix is to stop treating $P\Delta V$ as a correction applied after the fact, and instead build it permanently into the definition of the quantity you track. Define

$$H \equiv U + PV.$$

Recall the first law for a closed system doing only PdV work, $dU = \delta Q - P\,dV$. At constant pressure this becomes $dU + P\,dV = \delta Q$, and since P is constant, $P\,dV = d(PV)$, so

$$d(U + PV) = \delta Q\Big|_P \quad \Longrightarrow \quad dH = \delta Q\Big|_P.$$

That last step, $P\,dV = d(PV)$ holding only because P is held fixed, is the entire mechanism. It is the one algebraic fact that lets the boundary work term be absorbed into a state function's definition rather than computed as a path-dependent correction on the side. The result: at constant pressure, and only at constant pressure, ΔH equals the heat exchanged directly, with no separate bookkeeping for boundary work required at all. H does not describe new physics that U was missing; it is U with the one correction you were always going to need, in the one condition you almost always find yourself in, pre-subtracted once and permanently into the definition itself.

This is exactly the same logical move behind $F = U - TS$ and $G = U - TS + PV$ from the upcoming potentials chapter, a Legendre transform, chosen not because U was missing information, but because a different combination of U with its own conjugate variables happens to answer a specific, recurring experimental question with no extra arithmetic. F was built to make constant-volume, constant-temperature bookkeeping trivial. G was built to make constant-pressure, constant-temperature bookkeeping trivial. H is built to make constant-pressure heat bookkeeping trivial. None of them contain new physical content beyond what is already in U, T, P, V, and S; they are all the same underlying energy, reshuffled into whichever combination happens to be the natural variable set for the constraint you actually have control over in the lab [67][22].

5 Thermodynamics Potentials

Before discussing the individual thermodynamic potentials, it is worth taking a step back and asking what they are, why they are useful, and where they are actually used [13, 14, 15, 16, 17, 18]. In particular, we will briefly explore their role in thermodynamic simulations, what these simulations are, and why they have become indispensable tools in modern science and engineering.

Thermodynamic potentials are rarely emphasized in undergraduate engineering courses beyond their mathematical definitions. At most, students learn to manipulate a few equations before moving on to more application-driven topics. However, these potentials become increasingly important at the graduate level and are fundamental in fields such as materials science, chemical engineering, condensed matter physics, and computational chemistry [19, 20, 21, 23, 24, 25, 26].

One of their most important applications is in thermodynamic simulations. These simulations predict how a material or chemical system behaves under different temperatures, pressures, and compositions without requiring every possibility to be tested experimentally. They are used to design new alloys, optimize manufacturing processes, predict phase diagrams, develop battery materials, and even aid in the discovery of entirely new materials. Rather than simply describing equilibrium, thermodynamic potentials provide the mathematical framework that determines which equilibrium state is energetically favorable under a given set of conditions.

In this section, we will develop the thermodynamic potentials from the internal energy and show why each potential naturally arises when a different set of variables is held fixed. Rather than presenting them as quantities to memorize, we will see that each potential is tailored to answer a specific class of physical problems.

5.1 Helmholtz Free Energy

Stated honestly, the second law is not a statement about a system at all. It is a statement about the universe:

$$dS_{\text{tot}} = dS_{\text{sys}} + dS_{\text{sur}} \geq 0. \tag{1}$$

That is correct, and it is also nearly useless in practice. No engineer wants to model the entire surroundings of a reactor just to decide whether a desired reaction (the system) will proceed. What thermodynamic potentials do, at their core, is repackage Eq. (1) into a criterion about the system alone.

Our starting point is the fundamental relation for a simple compressible system,

$$dU = T\,dS - P\,dV + \mu\,dN, \tag{2}$$

which tells us that the natural variables of U are (S, V, N), where N is the number of particles and μ is the chemical potential. You see the trouble? Nobody has an entropy dial. Laboratories and process plants control temperature (via a thermostat, a furnace, a water bath) and pressure (via the atmosphere or a compressor), which are precisely the conjugate variables to S and V. We know the wrong half of each conjugate pair. Therefore we utilize the Legendre transform to change our dials.

A Brief Interlude: What a Legendre Transform Actually Does

Before doing this to U, it is worth understanding the trick in the abstract, because it is genuinely simple and it is easy to let the notation make it look more mysterious than it is.

Think of any function $f(x)$ as a curve. There are two completely equivalent ways to describe that curve. The first is the obvious one: list, for every x, the height $f(x)$. The second is stranger but just as complete: for every possible slope p, record the height of the single tangent line with that slope where it crosses the vertical axis.

$$f(x) \to f(x(p))$$

However you would soon notice that you would be loosing information if you simply translated the same $f(x)$ in any given direction. So what the transform does is to simply re-index the same curve, from "height as a function of position" to "intercept g as a function of slope."

$$g(p) = px(p) - f(x(p)) \qquad p \equiv \frac{\mathrm{d}f}{\mathrm{d}x}$$

This transform, however, only works for convex functions where slope $p(x)$ is unique. Nothing about the curve changed. Only the coordinate you use to label points on it changed, from x itself to the curve's own local slope at x. This is essentially it.
This matters here because internal energy has exactly this structure. $U(S, V, N)$ is a function of entropy, and its slope with respect to entropy is temperature, $T = (\partial U/\partial S)_{V,N}$. Entropy is a perfectly good variable mathematically, but it is a terrible one experimentally. Temperature, its slope, is something a thermostat controls directly. A Legendre transform is precisely the tool that lets us stop describing the system by its entropy and start describing it by its temperature instead, without losing a single bit of physical content in the process.

5.1.1 Applying the Transform to U

Now lets do this explicitly; however I do recommend you watching [68] beforehand to grasp this better as I will speed through the transformation. Using the recipe from the interlude with $f \longrightarrow U$, $x \longrightarrow S$, $p \longrightarrow T = (\partial U/\partial S)_{V,N}$, the raw Legendre transform of U with respect to S is

$$U^*(T, V, N) \equiv \big[\, TS - U \,\big]_{S=S(T,V,N)},$$

where $S(T, V, N)$ comes from inverting $T = T(S, V, N)$ at fixed V, N. To prove that we have changed the independent variables, differentiate the above directly using the ordinary product rule on TS:

$$\mathrm{d}U^* = T\,\mathrm{d}S + S\,\mathrm{d}T - \mathrm{d}U.$$

Substitute the fundamental relation, Eq. (2), for $\mathrm{d}U$:

$$\mathrm{d}U^* = T\,\mathrm{d}S + S\,\mathrm{d}T - \big(T\,\mathrm{d}S - P\,\mathrm{d}V + \mu\,\mathrm{d}N\big) = S\,\mathrm{d}T + P\,\mathrm{d}V - \mu\,\mathrm{d}N.$$

The two $T\,\mathrm{d}S$ terms cancel exactly. This cancellation is not a coincidence, it is the entire mechanism of the transform, the whole reason it removes S from the list of natural variables. What survives is a differential written purely in (T, V, N), exactly as promised: $(\partial U^*/\partial T)_{V,N} = S$, $(\partial U^*/\partial V)_{T,N} = P$, $(\partial U^*/\partial N)_{T,V} = -\mu$.

U^* already contains everything we set out to get, entropy has been fully swapped for temperature. But thermodynamicists don't usually work with U^* directly; they define the

Helmholtz free energy as its negative [10],

$$\boxed{F \equiv -U^* = U - TS} \tag{3}$$

This flip is a deliberate choice, not a mathematical requirement, and it's worth knowing why it's made. U itself has a familiar property: at fixed S, V, N, a system relaxes toward the state of lowest U, the same intuition as a ball settling to the bottom of a bowl. The raw transform $U^* = TS - U$ is the negative of U in exactly the sense that matters for this property, so if we used U^* as our potential, equilibrium at fixed T, V, N would show up as a maximum instead, mathematically equivalent, but a needless inversion of an intuition we'd rather keep. Defining $F = -U^* = U - TS$ restores the familiar "equilibrium is a minimum" picture, so everything you already know about how U behaves carries over directly to F, just in a different pair of coordinates. (Subsequent chapter proves this minimization explicitly; for now only the bookkeeping matters.)

Differentiating Eq. (3) directly, the sign flip carries through every term:

$$\mathrm{d}F = \mathrm{d}U - T\,\mathrm{d}S - S\,\mathrm{d}T = -S\,\mathrm{d}T - P\,\mathrm{d}V + \mu\,\mathrm{d}N. \tag{4}$$

The $T\,\mathrm{d}S$ term has been surgically removed and replaced by $-S\,\mathrm{d}T$. The natural variables of F are now (T, V, N): temperature, which we control with a thermostat, and volume, which we control with a rigid container. Nothing has been added to or subtracted from the physics; F contains exactly the same information as U, written in coordinates we can actually reach.[1]

5.1.2 F is minimized: the reservoir argument

Consider a system held at fixed volume and particle number, in thermal contact with a large reservoir at temperature T. The reservoir is large enough that its temperature does not change, so when the system absorbs heat δQ the reservoir's entropy changes by $\mathrm{d}S_{\mathrm{res}} = -\delta Q/T$. Since the walls are rigid, no work is done and the first law gives $\delta Q = \mathrm{d}U$. Substituting into Eq. (1),

$$\mathrm{d}S - \frac{\mathrm{d}U}{T} \geq 0 \quad \Longrightarrow \quad \mathrm{d}U - T\,\mathrm{d}S \leq 0. \tag{5}$$

At constant T the left-hand side is exactly $\mathrm{d}F$, so

$$\boxed{(\mathrm{d}F)_{T,V,N} \leq 0} \tag{6}$$

with equality at equilibrium.

[1] The transform is information-preserving precisely because U is convex in S. If it were not, the map $S \leftrightarrow T$ would not be one-to-one and the transform would lose information — which is why the convexity/stability conditions of Sec. 5.1.5 are not a technicality but a prerequisite.

It is worth being precise about what this inequality is saying, because it describes two different regimes at once. Away from equilibrium, any spontaneous change the system undergoes, a chemical reaction proceeding, heat redistributing internally, a phase converting, satisfies $dF < 0$ strictly: the free energy drops. The system does not evolve at random; among the changes available to it, the ones that actually occur are the ones that lower F. This is a statement about the direction of spontaneous change, not a static property. That process cannot continue forever, because F cannot decrease without bound. As the system evolves, it eventually exhausts every available change that would lower F further. At that point $dF = 0$: no accessible variation in the system's internal configuration changes F. This is the minimum.

One consequence worth stating explicitly: since F can only decrease or stay constant at fixed T, V, N, the system can never spontaneously leave its equilibrium once it is reached. Any fluctuation that raised F would violate Eq. (6), so the minimum is not just where the system ends up, it is where it stays.

It is worth pausing on the structure of $F = U - TS$, because it is the origin of nearly every competition in thermodynamics. A closed system held at fixed temperature T and volume V (in contact with a large thermal reservoir) does not minimize its energy, and it does not maximize its entropy either — it minimizes the combination $F = U - TS$. That combination is a compromise between two competing tendencies: the drive toward low-energy configurations (tight bonds, aligned spins, dense packing) and the drive toward high-multiplicity configurations (many accessible microstates). Temperature sets the exchange rate between the two, converting a change in entropy into an equivalent change in energy: a state that gains ΔS is worth pursuing energetically up to a cost of $T\Delta S$. At low T that exchange rate is small, so even a modest energetic advantage outweighs a large entropic one, and systems order. At high T the exchange rate is large, so even a modest entropic advantage outweighs a large energetic one, and systems disorder. Melting, magnetic transitions, polymer denaturation, and vacancy formation in crystals are all this single trade-off wearing different costumes. To build intuition for this, I highly recommend [69].

What does it actually mean for U to "win"? It means the system settles into a small number of low-energy configurations — strong bonds, tight packing, aligned spins — because the energetic cost of being anywhere else outweighs whatever is gained in multiplicity. Water freezing below 0°C is the clearest everyday example: liquid water has vastly more accessible microstates than ice (each molecule can translate and reorient almost freely), and yet the system still freezes, because forming a fully hydrogen-bonded lattice lowers U by more than $T\Delta S$ costs at that temperature. A crystal in general sacrifices enormous combinatorial freedom — there are vastly more disordered arrangements than ordered ones — in exchange for occupying deep energy wells. U wins whenever the energy saved by ordering exceeds T times the entropy given up.

What does it mean for S to "win"? It means the system abandons low-energy configurations in favor of whichever macrostate has overwhelmingly more accessible microstates, even though each individual microstate is less energetically favorable. This is not some mysterious force pulling the system toward chaos; it is simply that the disordered macrostate is so combinatorially vast that the system is overwhelmingly likely to be found there — the same reason a shuffled deck stays shuffled. Ice melting above 0°C is the same competition as above with the outcome reversed by temperature: past the melting point, $T\Delta S$ for melting exceeds the energy cost of breaking the lattice, so the liquid wins. Rubber elasticity is a more surprising example of entropy winning almost by itself: stretching a rubber band straightens its polymer chains into a small number of extended configurations, which is entropically expensive; released, the band snaps back not because of any restoring energetic force (for an ideal rubber, $\Delta U \approx 0$), but because the coiled, high-entropy state is simply the overwhelmingly more probable one.

5.1.3 F as available work

F is not just an abstract bookkeeping quantity, it measures how much useful work the system can still give up. Consider a closed system in contact with a reservoir at temperature T: the first law says $dU = \delta Q - \delta W$, where δW is the work done by the system on its surroundings. The second law bounds the heat it can absorb from the reservoir: $\delta Q \leq T\,dS$, with equality only for reversible processes. Combining the two,

$$\delta W \;\leq\; T\,dS - dU \;=\; -dF.$$

So the work extracted in any process is bounded above by $-\Delta F$, the drop in free energy, and a reversible process saturates that bound exactly. This is the literal meaning of "free" energy (historically denoted A, from the German Arbeit, work): U is the total energy stored in the system, but not all of it is available to do work, because the second law requires that some minimum amount, TS, be surrendered to the surroundings as heat just to keep the entropy books balanced. F is the leftover piece — the part of U that is free to be converted into work. Equilibrium, where F is minimized, is the state where the system has already given up every last bit of extractable work at that temperature; there is nothing more to gain, which is exactly why $dF = 0$ there. This is also why the balance between U and S matters so much: a system that is high in U but low in S relative to equilibrium still has "spare" free energy to release, and it will keep evolving, releasing work or heat, until that surplus is used up.

Rubber is the cleanest illustration. Stretching an ideal elastomer barely changes U at all (the covalent bonds are not being strained rather its coiled polymer chains unbending) but it dramatically reduces the number of chain conformations, so S drops. The restoring force is then almost purely entropic, $f = (\partial F/\partial(\text{Length}))_T \approx -T(\partial S/\partial L)_T$, which immediately predicts the experimentally famous result that a stretched rubber band under load contracts when heated (Gough–Joule effect), unlike essentially every other material.

5.1.4 Simulation

The Helmholtz free energy is the single most important potential for anyone doing microscopic modelling, for a very practical reason: it is the potential that matches the box you actually simulate.

When you run a molecular dynamics or Monte Carlo simulation, whether you're modelling a defect in a crystal, a polymer chain, or a fluid, you build a simulation cell of a fixed size, put a fixed number of atoms in it, and couple it to a thermostat that holds the temperature steady. You have, by construction, fixed V, fixed N, and fixed T. You did not fix the pressure. Pressure is not an input to the simulation, it is an output: you measure it afterward from the forces on the box walls. This is the opposite of a beaker on a lab bench, where the atmosphere pins the pressure and the volume is free to change as the reaction proceeds. Your simulation box is sealed and rigid; the beaker is open and floppy. Different constraints, different natural potential. G, as you will see shortly, is the free energy that answers "what's stable at fixed pressure," which is the chemist's question. F is the free energy that answers "what's stable at fixed volume," which is the modeller's question, because that's the constraint you actually imposed when you built the box.

So in practice, when you compute a defect formation energy, a stacking fault energy, a solvation free energy, or which of two crystal structures is more stable at a given temperature, what you are computing, whether the software calls it that or not, is a difference in F. The standard simulation techniques for this (thermodynamic integration [40], free energy perturbation [41], umbrella sampling [42]) all work by accumulating ΔF between two states, because F is the quantity that is actually well defined and computable at fixed T, V, N. If someone hands you a ΔU from a simulation and calls it a stability prediction, be suspicious; entropy differences between competing structures, vibrational, configurational, are frequently the deciding factor, and ΔU alone throws that information away.

There is a deeper reason this works, and it's worth knowing even if you never touch it directly: every microscopic simulation is implicitly sampling from the same statistical weighting, e^{-E_i/k_BT}, that defines the canonical partition function Z, and F is related to that partition function by the strikingly simple formula $F = -k_B T \ln Z$ [70]. Practically, this means F is not just a quantity you can compute from a simulation, it is the quantity your simulation's statistics are already built around, which is why the standard free-energy methods converge with far less sampling effort than trying to separately extract U and S and subtract. You rarely compute Z directly (it's an astronomically large sum), but every free-energy method in your simulation toolkit is, under the hood, a clever way of estimating $\ln Z$-differences without ever writing Z down explicitly.

The practical takeaway for an engineer: if you're asking a question at fixed pressure, phase stability of a material in service, which way a reaction runs on the bench, use G. If you're asking a question inside a simulation box, fixed volume, fixed atom count, use F, and recognize that most of the "free energy" language you'll see in computational materials science and molecular simulation papers is F, not G, even when the paper is silent about which one it means.

The Maxwell relation, and why it earns its keep

Start from a problem you've actually had: you need the entropy change of a fluid across a compressor or turbine stage, so you can compute an isentropic efficiency, or check how far a real process departs from an ideal one. You have a pressure transducer, a thermocouple, and a flow meter on the line. None of those measures entropy, because there is no such thing as an entropy meter, you cannot instrument a pipe for S the way you instrument it for P, T, or $\dot{V}$. So how does every process simulator on the market report an entropy value for the stream in that pipe?

Because $\mathrm{d}F$ is an exact differential, its mixed second derivatives must commute [11], and working through what that means for $F(T, V)$ gives

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V . \tag{7}$$

Read the right-hand side first: it says, if you know how a fluid's pressure responds to a small temperature change at fixed volume, something you can get directly from any P-V-T equation of state (ideal gas, van der Waals, Peng–Robinson, whatever correlation your simulator has loaded), then you automatically know how its entropy responds to a small volume change at fixed temperature. The left side is the unmeasurable quantity you wanted. The right side is built entirely from things you can look up or compute from an equation of state. Eq. (7) is the conversion factor between them.

This is not a one-off trick, it is the mechanism. Integrate Eq. (7) from ideal-gas conditions out to real operating conditions and you get a departure function: the correction that takes an entropy (or enthalpy, or Gibbs energy) computed from simple ideal-gas relations and corrects it for real-fluid, non-ideal behavior at the actual P and T of your process. Every steam table, every refrigerant property chart, every entropy or enthalpy value your process simulator hands you for a real fluid was generated this way: measure P-V-T behavior (cheap, direct, routine), then use a Maxwell relation to convert that into entropy and energy data (impossible to measure directly). If you've ever pulled an entropy value off a steam table and used it to size a turbine, you were, without knowing it, relying on Eq. (7) having already been integrated on your behalf.

5.1.5 Stability

One more result falls out of the same minimum-F argument, and this one you can sanity-check against your own intuition for how materials behave. Because equilibrium is a genuine minimum of F, not just a stationary point, the curvature of F at that minimum has to be non-negative in every direction. Translated into measurable quantities, that curvature condition says

$$C_V = -T\left(\frac{\partial^2 F}{\partial T^2}\right)_V \geq 0, \qquad \kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \geq 0. \tag{8}$$

C_V you already know: it's the heat capacity at constant volume, how much a material's temperature rises for a given amount of heat you dump into it. κ_T, the isothermal compressibility, is its mechanical twin: it measures how much a material's volume shrinks when you squeeze it harder at fixed temperature, how "squishable" it is. A large κ_T means a gas, soft, easily compressed. A tiny κ_T means a solid, stiff, barely budges no matter how hard you push. Both quantities are, in everyday terms, just a material's response to being disturbed, one thermally, one mechanically.

$$C_V = -T\left(\frac{\partial^2 F}{\partial T^2}\right)_V \geq 0, \qquad \kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \geq 0. \tag{9}$$

Think about what it would mean for either of these to go negative. A negative C_V would mean a material that gets colder as you add heat to it. A negative κ_T would mean a material that expands when you squeeze it harder. Neither of those describes any homogeneous material you have ever handled, and the stability condition is the formal statement of why: a chunk of material behaving that way would not sit still. The tiniest fluctuation, a slightly hotter patch, a slightly denser patch, would spontaneously grow instead of dying out, because growing it would lower F further. A true equilibrium phase can't do that; if it could, it wouldn't be at a minimum, and it would keep changing until it found one.

The cleanest place to watch $\kappa_T < 0$ actually show up is a simple, textbook model of a real gas, the van der Waals equation of state,

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT, \tag{10}$$

a small correction to the ideal gas law that accounts for two things real molecules do that ideal-gas molecules don't: they take up a bit of space of their own (the b term) and they weakly attract each other from a distance (the a term). An isotherm is simply what you get by picking one fixed temperature and plotting how P varies with V at that temperature, a single curve, one for each T you choose.

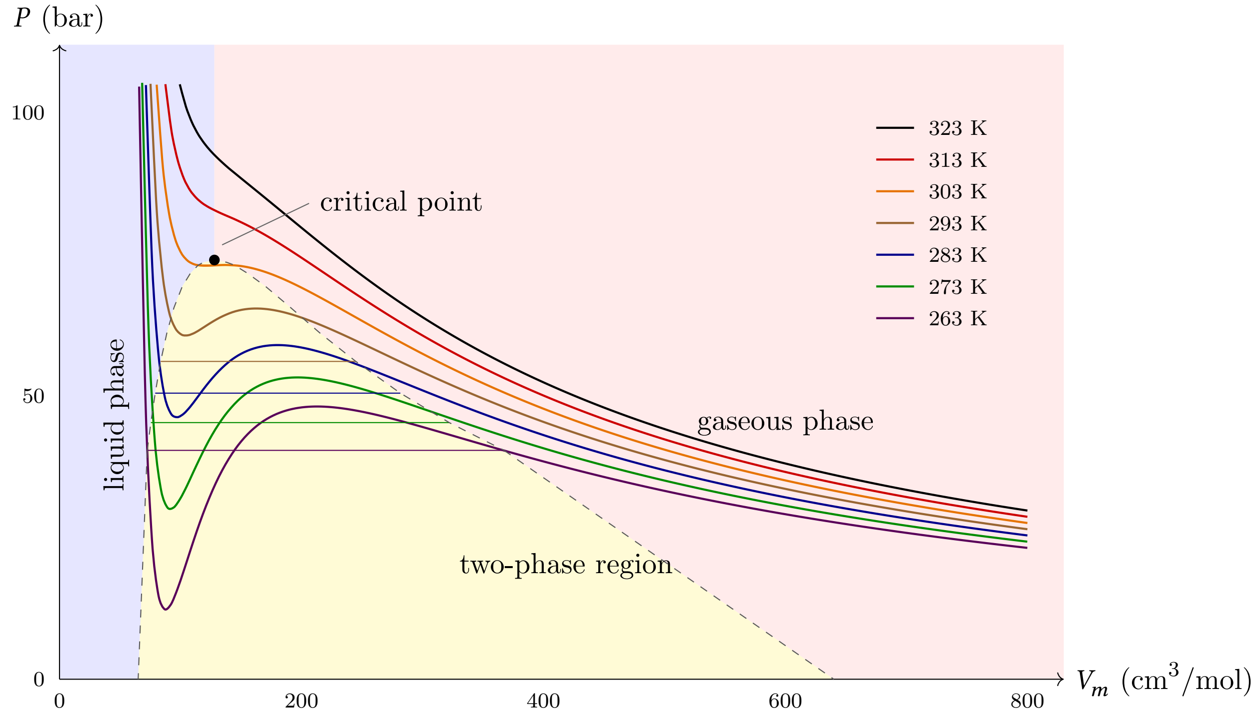


Figure 2: Van der Waals isotherms for CO_2 ($a = 3.640 \times 10^6$ bar cm^6 mol^{-2}, $b = 42.67$ cm^3 mol^{-1}), plotted at seven temperatures spanning the critical point, $T_c \approx 304$ K, $P_c \approx 74$ bar, $V_{m,c} = 3b \approx 128$ cm^3/mol. Below T_c, each isotherm develops the unstable S-shaped loop discussed in the text; the horizontal tie lines are the true Maxwell equal-area construction for each temperature, not an approximation, and mark where the fluid actually splits into coexisting liquid and vapor. The dashed curve is the binodal, the boundary of the two-phase region, meeting itself smoothly at the critical point.

At high temperature, this curve looks exactly like you'd expect: P falls smoothly and monotonically as V increases, the same shape the ideal gas law would give you. But pick a temperature below the fluid's critical temperature, and something strange happens: the curve stops being monotonic. Instead of falling smoothly, it dips down, turns around and rises, then turns around again and resumes falling, an S-shaped wiggle sitting in the middle of an otherwise ordinary-looking curve.

Here's the physical reason that wiggle appears at all. At small V, the molecules are packed close together, and the b term dominates: squeezing the fluid further means fighting real physical crowding, so P shoots up steeply as V shrinks, exactly like compressing a liquid. At large V, the molecules are far apart, attraction barely matters, and the curve behaves almost exactly like an ideal gas, P falling gently as V grows. The trouble is the middle ground, moderate density, where the attractive a term is strong enough to fight back against the pressure in a way the model can't resolve smoothly: the equation, doing exactly what its algebra demands, produces a stretch where P actually increases as V increases, exactly the reverse of every real material's behavior.

That reversed stretch is precisely the stability condition failing: $(\partial P/\partial V)_T > 0$ there is identical to $\kappa_T < 0$. The model isn't describing some exotic material that really expands when squeezed; it's telling you, in the clearest way an equation can, that no homogeneous single phase is allowed to exist at those conditions at all. Physically, the fluid resolves the contradiction the only way it can: instead of sitting on the unstable middle branch, it splits into two coexisting phases, liquid and vapor, sitting side by side at the same T and P, exactly what you already know happens inside a boiler drum or a refrigerant expansion valve. The wiggle in the math is the equation of state trying to draw you a phase boundary, without ever being told to.

Use F when: the system is thermostatted at fixed volume; you are working from a microscopic model or a partition function; you are running canonical (NVT) molecular dynamics or Monte Carlo; you want the maximum work from an isothermal process; you are dealing with magnetic, elastic, or lattice systems where volume is naturally the control variable.

5.2 Gibbs Free Energy

Fixed volume is convenient for a simulation box and inconvenient for almost everything else. Real chemistry happens in beakers, reactors, furnaces, and living cells, all of which sit open to a fixed ambient pressure and change volume freely. A process flow diagram never specifies the volume of a reactor as a design input, it specifies temperature and pressure, because those are the two things a plant operator actually controls with a valve and a heater. If F is the free energy for a sealed box, G is the free energy for exactly the situation you're in on a real process line: open, at atmospheric or set pressure, exchanging both heat and volume with the surroundings.

We build G the same way we built H from U not because F is missing physics, but because holding pressure fixed instead of volume introduces its own bookkeeping problem, and G is defined to pre-solve it. Perform the second Legendre transform, trading V for P:

$$\boxed{G \equiv U - TS + PV = H - TS = F + PV} \tag{11}$$

$$dG = -S\, dT + V\, dP + \sum_i \mu_i\, dN_i. \tag{12}$$

The natural variables of G are (T, P, N_i), the three things a chemical engineer specifies on a process flow diagram, and precisely the three things a materials scientist specifies when asking what phases a given alloy will contain. This is why G, not U, is the potential that appears throughout chemistry and metallurgy.

Repeating the reservoir argument for a system coupled to both a thermal reservoir at T and a pressure reservoir at P gives the corresponding equilibrium criterion,

$$\boxed{(dG)_{T,P} \leq 0} \tag{13}$$

with the same reading as for F: strictly negative while the system is still changing, zero once it has found the configuration, the phase, the extent of reaction, the composition, that no further change can improve on. The extra PV term in Eq. (11) is exactly the work the system must do to push the atmosphere out of the way in order to occupy its volume. A reaction that releases energy but expands while doing so does not get to keep all of that energy as useful output; some of it is spent shoving the atmosphere aside, and G is what remains after that cost is paid. This is the reason a reaction's heat release (ΔH) and its actual driving force (ΔG) are different numbers, and why engineers size equipment, and predict whether a reaction runs forward at all, using ΔG, never ΔH alone.

5.2.1 G is the chemical potential

Here is the single most useful fact in this whole section: G per mole is the chemical potential, $\mu = g$.

A Brief Interlude: What Chemical Potential Actually Is

Open a bottle of perfume in the corner of a room and within a few minutes you can smell it across the room. Drop a sugar cube into tea and, without stirring, the sweetness eventually reaches every corner of the cup. Leave a puddle on the pavement on a summer day and by evening it is gone, into the air. None of these things had a fan, a pump, or a person moving the molecules along. In every case, the molecules moved themselves, spontaneously, in one direction and never the other.

You already know the pattern behind two other flows. Heat moves from hot to cold, never the reverse, that pattern is temperature. A gas expands into a vacuum, never the reverse, that pattern is pressure. Matter follows a third pattern, just as automatic, just as blind, and everything in this chapter has secretly been about it already: it is G, seen from a new angle.

Every phase diagram, every reaction direction, every fuel cell voltage you've computed so far in this chapter came from G finding its minimum for the system as a whole. Chemical potential, μ, is what you get when you ask that same question one particle at a time: not "what's the lowest G this whole system can reach," but "what does it cost, or gain, in G, to add one more particle of this substance, right here, right now." Formally,

$$\mu_i = \left(\frac{\partial G}{\partial N_i}\right)_{T,P,N_{j\neq i}} = \left(\frac{\partial U}{\partial N_i}\right)_{S,V,N_{j\neq i}}.$$

Don't let the derivative distract from the idea underneath it. μ is not a new player walking onto the stage, it is G's own fine print, the same ledger that decided whether a fuel cell had voltage to give, or whether an alloy split into two phases, now kept separately for each substance, in each place it happens to be. And wherever a substance is free to move, by diffusing, evaporating, dissolving, reacting, or crossing a membrane, it moves from where μ is high to where μ is low, and keeps moving until μ is equal everywhere it has access to. Nothing is pushing it there in the Newtonian sense; μ is simply telling you which direction is downhill for matter, the same way altitude tells you which way water will flow without you ever computing the force on each molecule of it.

Once you have this picture, you start seeing μ, and behind it G, everywhere, not just in a beaker. Salt melts ice on a winter road because dissolved salt lowers μ for water in that mixture below μ for solid ice, so the ice, chasing equilibrium the only way it knows how, melts into it. You sweat to cool down because evaporation is water moving from high μ (liquid on your skin) to low μ (vapor in the dry air around you), and it carries heat away as it goes. A steel bridge corrodes, slowly and invisibly, over decades, because bare iron sits at a higher μ than iron oxide does in the presence of water and oxygen, and the metal is simply doing what all matter does, sliding toward the lower value, one atom at a time. A battery produces current because the (electro)chemical potential of electrons is higher at one electrode than the other, and current is nothing more than μ finding its way to level.

None of these needed a person to explain "why." They happen on their own, driven by the same single quantity, the same way a ball rolls downhill without anyone computing its gravitational potential energy first. Once you can see μ, you stop asking why something dissolved, corroded, evaporated, or diffused as though each were its own separate mystery, and start recognizing all of them as the same question, asked by nature over and over: where is μ still uneven, and how fast can it be evened out. And every time you catch yourself asking that question, about a road, a bridge, a battery, a cup of tea, you are asking nothing more than where G still has room to fall. That question, asked at every scale from a teacup to a bridge truss to a battery cell, is the entire content of chemical equilibrium, and you have been carrying the tool to answer it since the moment you understood what G was minimizing toward.

Before writing down the formal result, it helps to see where it comes from. Imagine two identical beakers of the same substance, each held at the same temperature T and pressure P, each containing N moles. Now pour one into the other. Nothing about the intensive state has changed — T and P are exactly what they were before, since both beakers were already at the same values. But every extensive quantity has exactly doubled: twice the entropy, twice the volume, twice the particles, and twice the Gibbs free energy. This is the defining property of an extensive function: at fixed T and P, G scales linearly with the amount of stuff. That single fact, formalized, is Euler's theorem for homogeneous functions, and it gives:

$$G = \sum_i \mu_i N_i, \qquad \text{so for one component} \qquad \mu = \frac{G}{N} = g. \tag{14}$$

Here $\mu_i \equiv (\partial G/\partial N_i)_{T,P,N_{j\neq i}}$ is the chemical potential of species i: the cost, in Gibbs free energy, of adding one more mole of species i to the system while holding T, P, and everything else fixed. For a single-component system this is just the molar Gibbs free energy, g, the free energy price tag attached to each mole of substance.

It's worth noticing why this identity is uniquely clean for G and has no equally simple counterpart for U or F. U and F are also extensive, and Euler's theorem applies to them too, but their natural variables — S, V, N for U; T, V, N for F — include extensive variables besides N (S or V). Pouring one beaker into the other doubles S and V as well as N, so the Euler sum for U has three surviving terms: $U = TS - PV + \mu N$. G's natural variables are T, P, and N, and T and P are intensive, they don't double when you combine the beakers, so they contribute nothing to the sum. Only the N term survives. That's the whole reason Eq. (14) is so simple: G is the one potential whose natural variables are "mostly intensive," so essentially all of it collapses into a single term, μN.

Practically, this is why every phase diagram you have ever read is built by comparing $g(T, P)$ curves, one per phase, and picking whichever is lowest at each condition: a system at fixed T, P minimizes G, and for a single-component phase, $G = Ng(T, P)$, so minimizing G at fixed N is the same as minimizing g per mole. Plot $g_{\text{ice}}(T, P)$ and $g_{\text{water}}(T, P)$ against each other, and whichever curve sits lower at a given (T, P) is the phase you actually observe; the crossing point of the two curves is the melting line.

It's also why phase diagrams have tie lines rather than single points in a two-phase region. Picture salt dissolving between a liquid phase and a solid crystal phase in contact: molecules of each species can cross the phase boundary in either direction, and the system settles down not when the two phases have the same composition (they generally don't) but when moving one more molecule of species i from one phase to the other costs nothing, i.e. when μ_i is equal in both phases. It's the same logic as prices equalizing across two connected markets: goods flow from the cheap side to the expensive side until the price is the same everywhere, not until the quantities on each side are equal. That equality of μ_i, not equality of composition, is the actual equilibrium condition, and a tie line just connects the two different compositions that happen to share the same μ_i for every species at a given T, P.

Differentiating Eq. (14) gives a second, independent expression for $\mathrm{d}G$ beyond the one in Eq. (12). Since both must be true simultaneously, subtracting one from the other cancels the $\mu_i\,\mathrm{d}N_i$ terms and leaves a relationship purely among the intensive variables, the Gibbs–Duhem relation [9]:

$$S\,\mathrm{d}T - V\,\mathrm{d}P + \sum_i N_i\,\mathrm{d}\mu_i = 0. \tag{15}$$

In plain terms: you don't get to choose T, P, and every μ_i independently. Fix all but one, and the last is pinned down by the rest.

5.2.2 Phase equilibrium and the Clausius–Clapeyron equation

Here's a question with obvious practical stakes: why does water boil at a lower temperature on a mountain, and why does a pressure cooker work at all? Both are really the same question in disguise: how does the boiling temperature shift when the pressure changes? The answer turns out to follow directly from the "lowest g wins" picture from the previous section, once you push it one step further.

Recall that at any fixed (T, P), the phase you observe is whichever one has the lower molar Gibbs energy g. Boiling is the special case where neither phase is preferred: liquid and vapor coexist at a given (T, P) exactly when their g curves cross, i.e. when

$$g_{\text{liq}}(T, P) = g_{\text{vap}}(T, P). \tag{16}$$

Away from this condition, one phase has strictly lower g and the other phase converts entirely into it: there's no such thing as boiling water at, say, room temperature and atmospheric pressure, because $g_{\text{liq}} < g_{\text{vap}}$ there and the liquid simply wins outright.

Now here is the key move: this crossing condition isn't satisfied at just one isolated point in the (T, P) plane, it's satisfied along an entire curve, the familiar liquid–vapor boundary line on a phase diagram. Raise the pressure, and the temperature at which g_{liq} and g_{vap} cross moves too, tracing out that boundary line as you go. That line is what we mean by the coexistence curve: the set of all (T, P) pairs at which liquid and vapor can sit side by side in equilibrium. It's the line a pressure cooker walks up as it pressurizes, and it's the reason a mountaintop, at lower atmospheric pressure, boils water at a lower temperature, you've moved to a different point on the same curve.

The reason this lets us calculate anything is that the defining condition, $g_{\text{liq}} = g_{\text{vap}}$, must hold not just at one point on the curve but at every point on it, by definition of what the curve is. So if you take two nearby points on the coexistence curve, separated by a small step $\mathrm{d}T$ and $\mathrm{d}P$, the equality has to survive that step too:

$$g_{\text{liq}} + \mathrm{d}g_{\text{liq}} \;=\; g_{\text{vap}} + \mathrm{d}g_{\text{vap}}. \tag{17}$$

Since $g_{\text{liq}} = g_{\text{vap}}$ to begin with, this collapses to a statement purely about the changes:

$$\mathrm{d}g_{\text{liq}} = \mathrm{d}g_{\text{vap}} \to -s_1\,\mathrm{d}T + v_1\,\mathrm{d}P = -s_2\,\mathrm{d}T + v_2\,\mathrm{d}P \quad\Longrightarrow\quad \frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta s}{\Delta v} = \frac{L}{T\,\Delta v}, \tag{18}$$

In words: whatever combination of $\mathrm{d}T$ and $\mathrm{d}P$ you choose, if you want to stay on the coexistence curve, the resulting change in g has to come out identical for both phases. That constraint is exactly what pins down the slope of the curve, $\mathrm{d}P/\mathrm{d}T$, and to extract it we expand $\mathrm{d}g_{\text{liq}}$ and $\mathrm{d}g_{\text{vap}}$ separately using Eq. (12).

where $L = T\Delta s$ is the latent heat. Three lines of algebra from the definition of G give the slope of every boundary on every phase diagram. This is not a specialty equation; it's the tool that generates every refrigerant pressure-temperature chart in an HVAC handbook, tells a process engineer how much a distillation column's operating pressure will shift a component's boiling point, and explains why a pressure cooker, by raising P, raises the boiling point of water and lets food cook faster. It also explains the one famous anomaly you've probably heard without knowing the reason: ice melts under pressure, because water is one of the few substances with $\Delta v < 0$ on melting (ice is less dense than liquid water), so its fusion line slopes backwards, colder temperatures require higher pressure to stay liquid, which is part of why a glacier can flow and why ice skates work better on some accounts of the effect than others.

5.2.3 Chemical reactions: the enthalpy–entropy contest

For a reaction running at fixed T and P, there's a single number that tells you everything you need to know about which way it wants to go: the reaction Gibbs energy, $\Delta_r G$. Think of it as a compass needle. If $\Delta_r G$ is negative, the reaction runs forward, reactants turning into products. If it's positive, it runs backward. If it's zero, the reaction has nothing left to gain either way and just sits at equilibrium. This is exactly the same U-versus-S competition as before, just relabeled: now ΔH (heat released or absorbed by breaking and forming bonds) plays the role of U, and ΔS (how the number and freedom of molecules changes) plays the role of entropy, with T again setting the exchange rate between them.

The genuinely interesting part is what this buys you. $\Delta_r G$ isn't just a signed number you measure after running the reaction; you can predict it before running the reaction, from tables of formation energies that chemists have already measured for thousands of compounds. That prediction tells you the equilibrium constant K, and K tells you how far the reaction will go, essentially the ratio of products to reactants once things settle down. In other words, someone can tell you the eventual yield of a reaction that has never been run in a lab, purely by looking up numbers for the ingredients separately. This is why chemical engineers can screen thousands of candidate reactions on paper before ever touching a beaker.

The part with real practical teeth is that $\Delta_r G = \Delta H - T\Delta S$ can change sign as T changes, because ΔH barely moves with temperature while the $T\Delta S$ term grows right along with it. A reaction that is comfortably favorable in the cold can become unfavorable in a furnace, or vice versa, purely because the entropy term picks up weight as T rises.

The Haber process, making ammonia from nitrogen and hydrogen gas ($N_2 + 3H_2 \rightarrow 2NH_3$), is a textbook example of this tension. The reaction releases heat ($\Delta H < 0$, energetically favorable), but it also takes four gas molecules and crams them into two, which is a big entropy loss ($\Delta S < 0$). At low temperature, the energy term dominates and the reaction is favorable, but running it cold makes it agonizingly slow (a kinetics problem, not a thermodynamics one). Run it hot enough to go at a useful rate, and the entropy penalty starts winning, dragging equilibrium back toward the reactants. The industrial process exists at all only because engineers found a temperature and pressure window, and added a catalyst, where both problems are tolerable at once, this is precisely why the Haber process runs at extreme pressure: squeezing the gas favors the side with fewer molecules, which fights back against the entropy penalty.

The same enthalpy-versus-entropy sign flip, at industrial scale, is the entire idea behind an Ellingham diagram [12], the tool a metallurgist uses to pick which reducing agent will strip oxygen out of a metal ore at a given furnace temperature. Carbon becomes a better reducing agent as the furnace gets hotter, not because carbon atoms get chemically stronger with heat, but because the reaction $2C + O_2 \rightarrow 2CO$ turns one gas molecule into two, gaining entropy, so its favorability keeps climbing with temperature even as competing reactions plateau or worsen. Above a crossover temperature, carbon's line dives below the metal oxide's line on the diagram, and reduction turns spontaneous. A blast furnace's operating temperature is, in effect, chosen by reading this crossover off a chart. It's a nice example of how a single thermodynamic idea, the entropy term outweighing the enthalpy term at high enough T, quietly decides something as concrete as how iron has been smelted for millennia.

5.2.4 Non-expansion work and the fuel cell

G answers a question every energy-conversion engineer eventually asks: of the energy a reaction releases, how much can actually be extracted as useful work, electrical, mechanical, shaft work, rather than just heat? For a process at fixed T and P, the bound is

$$W_{\text{non-}PV,\text{ by}} \leq -\Delta G, \tag{19}$$

that is, $-\Delta G$ caps the electrical, mechanical, or chemical work extractable after the atmospheric pushing work has already been paid for out of ΔH. This is the design ceiling for anything that converts chemical energy directly into work without going through a heat engine: batteries, fuel cells, electrolyzers, muscle. For an electrochemical cell this becomes $\Delta G = -nFE$, the Nernst relation, directly linking a thermodynamic quantity to a number you can read off a voltmeter, and it produces one of the more striking numbers in energy engineering. For $H_2 + \frac{1}{2}O_2 \rightarrow H_2O(\ell)$ at standard conditions, $\Delta H^\circ \approx -286\ \text{kJ}\,\text{mol}^{-1}$ and $\Delta G^\circ \approx -237\ \text{kJ}\,\text{mol}^{-1}$, giving a reversible cell voltage of

$$E^\circ = \frac{237{,}000}{2 \times 96485} \approx 1.23\ \text{V} \tag{20}$$

and a thermodynamic efficiency ceiling of $\Delta G/\Delta H \approx 83\%$. This is worth sitting with: a fuel cell is not bound by the Carnot limit, it is not a heat engine and never converts heat to work through a temperature difference, so a Carnot-efficiency argument does not apply to it at all. But it is still bound, by ΔG, and the missing 17% is not a manufacturing defect that better engineering will someday close. It is $T\Delta S$, the entropy of the reaction, which must leave the system as heat by the second law, no matter how well the cell is built. Knowing which inefficiencies are fundamental (this one) versus which are recoverable (activation losses, ohmic losses, mass transport losses, all the things that make a real fuel cell fall well short of even 83%) is exactly the distinction G draws for you before you've built anything.

5.2.5 The Gibbs Maxwell relation

Exactness of Eq. (12) gives

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P = -V\beta, \tag{21}$$

where β is the thermal expansion coefficient, again something directly measurable, unlike the entropy derivative on the left. This one relation is the reason a gas can cool when you let it expand through a valve with no moving parts and no heat removed. The Joule–Thomson effect is important in throttling-based refrigeration and many gas-liquefaction processes.

Whether a given gas cools or heats on throttling depends on the sign of a quantity built from Eq. (21) compared to C_P, which is why some gases (most, at room temperature) cool when throttled and are useful for refrigeration, while others (like hydrogen and helium, at room temperature) actually heat up when throttled and must first be pre-cooled below their inversion temperature before expansion will liquefy them, a fact that is not optional trivia if you are designing a hydrogen or helium liquefaction plant, it is the reason the plant's process flow diagram has an extra pre-cooling stage that a nitrogen plant doesn't need.

Table 1: The four common potentials. Each is a Legendre transform of U, obtained by trading an extensive natural variable for the intensive one you can actually control.

Potential	Definition	Differential	Natural vars.	Minimized at fixed
U	—	$T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$	S, V, N	S, V, N
H	$U + PV$	$T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N$	S, P, N	S, P, N
F	$U - TS$	$-S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N$	T, V, N	T, V, N
G	$U - TS + PV$	$-S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N$	T, P, N	T, P, N

The pattern across all four rows is the same one you've now seen twice: each potential is U repackaged so that the quantity held fixed in your actual experiment (a rigid box, an open beaker, a thermostatted simulation, a bench reaction) is exactly the potential's natural variable, which is what makes that potential the one that's minimized, the one whose derivatives give you measurable quantities for free, and the one whose sign tells you which way the system is headed.

Use G when: the system is open to the atmosphere or held at fixed pressure; you are computing phase diagrams, solubilities, or reaction equilibria; you need cell voltages or electrochemical driving forces; you are asking which of several competing phases or products a system will actually choose.

5.3 Thermodynamic Simulations

5.3.1 What the computer is actually doing

Imagine you're handed a fixed bag of atoms, say iron and carbon, and told: at this temperature and this pressure, arrange these atoms however you like, split them into a solid here, a liquid there, two different crystal structures if you want, in any proportion. Which arrangement does nature actually pick? That question, asked at every temperature and composition, is a phase diagram. And it has one answer, because we already know the rule from earlier in this chapter: at fixed T and P, nature picks whichever arrangement has the lowest total Gibbs energy.

$$\min_{\{n_i^\alpha\}} G_{\text{total}}(T, P, \{n_i^\alpha\}) = \min \sum_\alpha \sum_i n_i^\alpha \mu_i^\alpha \quad \text{subject to} \quad \sum_\alpha n_i^\alpha = n_i^{\text{tot}}, \tag{22}$$

Don't let the double sum obscure how simple the idea is: each candidate phase α quotes a "price per atom" (μ_i^α) for holding each species i, and the computer just tries every way of splitting the atoms among the candidate phases and keeps whichever split has the lowest total bill. Nothing about which phases show up, in what amounts, with what compositions, is assumed in advance — it simply falls out of the minimization [37, 38]. Sweep this calculation across a grid of temperature and composition, and wherever the winner changes, you've just drawn a phase boundary. A phase diagram, in this light, is nothing more than a map of who wins Eq. (22) at each point.

The catch is that you can't know a phase loses unless you know what it would have cost to win. That means you need a free energy formula for every candidate phase, including ones that are hypothetical or metastable in the region you care about. You cannot rule out BCC iron without knowing what BCC iron's free energy would have been, even in a regime where it never actually appears.

5.3.2 CALPHAD: fitting free energy surfaces

So where do those free energy formulas come from? The honest answer, for most of the periodic table, is: they're fitted, not derived. This is the CALPHAD (CALculation of PHAse Diagrams) approach [29, 27, 28, 31, 32, 33, 34, 35], and the logic behind it is closer to economics than physics. You can measure, in a lab, how two or three elements behave when mixed, their melting points, their phase boundaries, how much heat they release. You cannot easily do the same for an eight-element superalloy; the experiment would take a career. So instead, you fit a flexible mathematical model to the two- and three-element systems you can measure, and trust that model to extrapolate sensibly into the eight-element mixtures industry actually casts. It's the same move as calibrating an economic model on a few well-studied countries and using it to forecast a market you've never observed directly.

The model itself is just three ingredients stacked together: the free energy of the pure components on their own, the entropy gain from mixing them (mixing always helps entropy, since it multiplies the number of possible arrangements), and a leftover "everything real" correction term that soaks up all the messy chemistry, fitted directly to data [30].

The payoff is considerable. These fitted databases, built from inexpensive binary and ternary measurements, are what let engineers predict brittle phases in stainless steel, design precipitation-hardened superalloys, and screen high-entropy alloy compositions, all before melting a single ingot. It's the same trick as predicting a chemical reaction's yield from tabulated data without running it, just extended to metallurgy.

A neat extension is the Scheil–Gulliver model, which adds a crude but useful kinetic assumption: as a casting solidifies, assume the solid never diffuses (atoms freeze in place once solid) but the liquid stays perfectly mixed. Picture freezing a pot of soup very slowly from the bottom up: each new layer that freezes locks in whatever composition happened to be left in the liquid at that instant, and never talks to the solid layers below it again. Repeatedly re-solving Eq. (22) as this freezing front advances gives you the segregation patterns and hot-tearing risk of a real casting.

5.3.3 Free energy from atoms: why it is hard

Here's a genuinely strange fact: a molecular dynamics simulation [39], which tracks every atom's position and velocity in exquisite detail, is <u>bad</u> at computing free energy, even though it's very good at computing ordinary energy. Why the difference?

Energy is something you can average over a trajectory: watch the atoms bounce around for a while, average what you see, and you have your answer. Free energy is not like this, because it depends on <u>every</u> configuration the system could possibly be in, including the ones it almost never visits. It's like trying to measure the size of a swimming pool by tracking where a swimmer spends their time. If they spend nearly all of it in the shallow end, watching them tells you a lot about the shallow end and almost nothing about the deep end, even though the deep end still counts toward the pool's total size. A simulation run the ordinary way spends its time exactly where the energy is low and misses the vast, improbable regions that still contribute to the free energy.

The standard fix is to stop trying to compute free energy directly, and instead compute a <u>difference</u> along an artificial path you construct on purpose. You build a slow, imaginary ramp connecting a system you already understand to the one you actually want, and add up the cost of turning that ramp's knob from one end to the other. Each individual step along the ramp is an ordinary average, the kind a simulation is good at, so the trick converts an impossible calculation into a sequence of easy ones. Umbrella sampling, metadynamics [44], and free energy perturbation are all variations on the same idea: deliberately force the simulation into the unlikely regions of the pool, then correct the bookkeeping for having cheated [43, 45].

This is not an academic exercise. It is literally how a drug company predicts, in software, whether a candidate molecule will bind tightly to a target protein, before synthesizing a single milligram of it.

A different route to the same destination comes from quantum mechanics directly: density functional theory (DFT) can compute a material's energy at absolute zero with no fitting at all [36], and adding in how the atoms vibrate (their phonon spectrum) extends that into a free energy that depends on temperature. This is the machinery behind the huge public materials databases that rank hypothetical new compounds by stability, the "energy above hull" number you'll see quoted in computational materials papers is a genuine free energy comparison, not a rule of thumb.

5.3.4 Beyond equilibrium: free energy functionals

Everything so far has answered "where does the system end up." But often the interesting part is the journey, not the destination: the branching shape of a snowflake, the striped pattern a cooling alloy segregates into, the uneven way lithium metal deposits inside a battery electrode. Phase field modeling captures this by writing the free energy not as a single number, but as something that depends on a field spread out over space, $\phi(\mathbf{r})$.

Picture ϕ as a wash of paint: zero wherever the material is liquid, one wherever it's solid, blurred smoothly across the boundary rather than switching instantly like a hard edge. That blurring is deliberate. The extra term in the energy that penalizes steep changes in ϕ is exactly what makes it costly to have an interface at all, which is just another name for surface tension. The system then evolves the same way a marble rolls downhill: at every point in space, ϕ shifts in whichever direction lowers the total free energy fastest.

Run this simulation and you get, essentially for free, the branching fingers of a solidifying metal casting, the speckled pattern of a mixture separating into two phases mid-motion, and the uneven plating that makes lithium batteries a safety concern. The same currency shows up in industrial process simulators like Aspen Plus and HYSYS: a "flash" calculation there is nothing more than Eq. (22) in disguise, equalizing chemical potential between a liquid and a vapor stream.

5.3.5 Practical cautions

Three things go wrong often enough to be worth saying plainly.

First, equilibrium is not kinetics. All of this machinery tells you where a system is headed, never how long the trip takes. Diamond is less stable than graphite at room conditions and will happily remain diamond for longer than anyone alive will notice. Every piece of hardened steel, every metallic glass, every gemstone on a ring is a structure that a purely equilibrium calculation would confidently predict shouldn't exist.

Second, the answer is only as good as the data it was fitted to. A solver is a very reliable machine for finding the bottom of whatever free energy surface you handed it. Fit that surface over a certain temperature range, then ask about a temperature far outside it, and the model will hand back a number with complete confidence, even though nothing guarantees it means anything. Extrapolating past the fitted range is the single most common way these tools get misused.

Third, a solver can't tell a real minimum from a merely nearby one. Real free energy landscapes for complicated alloys are bumpy, with many local dips, and a solver started from a bad initial guess can settle into one of the wrong ones and report it with the same confidence it would give a correct answer. Good software fights this by sampling many starting points, but a surprising result is always worth treating as a hypothesis to check, not a fact to trust.

None of this undercuts the main point. Thermodynamic potentials are what let an engineer ask "what will this alloy do at 900°C" and get a defensible answer before the furnace is ever switched on. They started out as a bookkeeping trick, a way to make the second law talk about the system instead of the whole universe, and ended up as the computational backbone of how modern materials and chemical processes get designed.

6 Diffusion Equation

Heat conduction arises from microscopic interactions in which thermally agitated molecules transfer energy to their neighbors through collisions and intermolecular forces. This process redistributes energy from regions of higher temperature to regions of lower temperature. When the process is time-dependent, the evolution of temperature in space closely resembles a diffusion phenomenon: thermal energy spreads down temperature gradients, analogous to how particles diffuse from regions of high concentration to low concentration. As a result, the governing equation for transient heat conduction has the same mathematical form as a diffusion equation (compare 23 and 24). Thus from here on now, the diffusion equation will be used synonymously with the transient heat conduction equation.[1]

$$\frac{\partial C}{\partial t} = D\nabla^2 C \tag{23}$$

This section will discuss the derivation of the heat diffusion equation only. This is because the diffusion equation is one of the small handful of partial differential equations in physics whose consequences reach far outside the problem that motivated it, and several of those consequences are genuinely surprising the first time you see them: the equation has a built-in clock that tells you how far something has spread without solving anything, it destroys information about initial conditions in a mathematically precise sense, it propagates disturbances at infinite speed even though nothing physical actually does, and its solution is quietly the same object that governs a random walk, a stock price, and the width of a doped region in a silicon wafer. Thus its implications, application and consequences carries enough content to be its own book. Thus, I encourage the reader to dive into the internet and lose themselves in the rabbit hole.

6.1 Derivation of the Heat Diffusion Equation

Lets focus our attention to a bar rod with the left side kept at T_1 and the right side kept at T_2 where $T_1 > T_2$. We have just created out isothermal boundary layers. We can easily discern that heat transfer will occur from the hot side to the cold side but we need to think more technical and adhere to a reductionist perspective. In a specified control surface, what can happen at the boundaries? This question is important in number of ways as energy crosses into or out of the system as heat transfer only at the boundaries and all boundaries must be considered to prevent any energy escaping from our radar (1st Law of Thermodynamics can not be violated under any circumstance).

We can think of the volume of the piece of rod from $x = a$ to $x = b$ as having some internal energy $H(x,t)$. However we know that this energy is subject to change as heat propagates through. This can be expressed as

$$\frac{d}{dt}\int_a^b H(x,t)\,dx$$

At the boundary $x = a$, heat enters the domain, while at $x = b$, heat leaves it. This transfer is described by a heat flux $J(x,t)$, which governs the rate at which energy crosses the boundaries and thereby affects the internal energy within the volume. In addition, there may be external sources (a candle) or sinks (wind) that transfer energy across the boundaries perpendicular to the x-direction (e.g., from the top and bottom surfaces), further contributing to changes in the internal energy. I will denote the external source as $S(x,t)$. Thus using the 1st law of thermodynamics we can write:

$$\frac{d}{dt}\int_a^b H(x,t)\,dx = J_x(a,t) - J_x(b,t) + \int_a^b S(x,t)\,dx$$

Which can be re-written using the fundamental rule of calculus as

$$\int_a^b \frac{\partial H(x,t)}{\partial t}\,dx = -\int_a^b \frac{\partial J_x(x,t)}{\partial x}dx + \int_a^b S(x,t)\,dx$$

This equation is very important as we can see that if we merge the terms on the left side under a single integral, we get a linear differential representing the local 1st law of thermodynamics in heat conduction.

$$\frac{\partial H(x,t)}{\partial t} + \frac{\partial J_x(x,t)}{\partial x} - S(x,t) = 0$$

And we already know what the internal energy is for a material, and the heat flux through a material.

$$H = \rho c_p T, \qquad J_x = -k\frac{\partial T}{\partial x} \quad \text{(from Fourier's Law)}$$

Where ρ is the density of the solid material, c_p is the specific heat capacity at constant pressure, and k is the thermal conductivity. For now we will consider when these physical constants are constant in time and space. We can now express this as:

$$\frac{\partial}{\partial t}[\rho c_p T] + \frac{\partial}{\partial x}\Big[-k\frac{\partial T}{\partial x}\Big] - S(x,t) = 0$$

$$\rho c_p \frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2} + S(x,t) \tag{24}$$

This is the heat diffusion equation for 1 dimensional, time dependent, constant physical coefficient heat conduction.

What this physically represents is how the temperature at a particular point evolves over time. In heat conduction terms, the internal energy changes due to heat flux and external/internal heat generation (or cooling). For properly defining a certain configuration, however, equation (24) requires two spatial boundary values and a time boundary value of the following form

$$T(x_1, t) = \alpha(x_1, t), \qquad T(x_2, t) = \beta(x_2, t), \qquad T(x, t_1) = \gamma(x, t_1),$$

It is also important to note that in 1D heat conduction the shape of the object, for example, rectangular bar, circular bar, triangular bar, does not matter but only the cross sectional surface area A_s does as heat transfer is defined with it. When this problem becomes 2D we will be concerned more about the shape.

7 Thermoelectric Devices

A thermoelectric device does something that sounds, at first hearing, like it should not be possible without moving parts: it converts a temperature difference directly into an electric current, or run in reverse, converts an electric current directly into a temperature difference. No turbine, no working fluid, no piston. A block of doped semiconductor with a hot side and a cold side is, quietly and without complaint, a heat engine.

This is not a niche curiosity. Thermoelectric generators power the Voyager and Cassini spacecraft, decades past their original mission timelines, off nothing but the steady decay heat of a plutonium pellet [57, 58]. Peltier coolers sit inside every portable wine cooler, every camping fridge, and increasingly, inside the packaging of high-power laser diodes and infrared detectors that need to be held at a fixed temperature to nanometre-level stability. Automotive waste-heat recovery, which tries to skim usable electricity off the roughly 60% of fuel energy that leaves a combustion engine as heat, is a thermoelectric materials problem before it is anything else.

What makes these devices worth a chapter of their own, in a paper built from thermodynamic fundamentals, is that they are among the cleanest physical demonstrations that heat and charge transport are not separate phenomena — they are coupled, and the coupling is not incidental but required by the same second law that gave us the potentials mentioned in the previous sections. We will build the three thermoelectric effects up from the coupled transport of charge and heat, show that they are not three independent effects but one effect seen from three angles, connect the whole framework back to entropy production, and finish with the figure of merit that determines whether any of this is any good as an engine.

7.1 The Seebeck Effect

In 1821, Thomas Seebeck noticed something odd: a loop made of two different metals, with one junction warm and the other cool, deflected a nearby compass needle [46]. No battery, no moving parts, just a temperature difference, and yet current was flowing. Take away the compass and instead measure the voltage across the open loop, and you get the modern statement of the effect: a temperature difference ΔT across a conductor produces a voltage,

$$\Delta V = -S\,\Delta T, \qquad \text{or differentially} \qquad \mathbf{E} = S\,\nabla T, \tag{25}$$

where S, the Seebeck coefficient, is typically tens to hundreds of μV/K, a small voltage, but a free one, generated from nothing but a hot side and a cold side.

The sign convention trips people up constantly, so it's worth fixing with a picture rather than a rule. In conventional single-carrier semiconductors, p-type materials generally have positive S, while n-type materials generally have negative S. Holes pile up on the cold side, leaving the hot side positively charged relative to it n-type semiconductors have $S < 0$; p-type have $S > 0$. This isn't bookkeeping trivia, it's the literal instruction a device designer follows to decide which doped material goes on which side of a thermoelectric module.

7.1.1 Where it comes from: carriers as a gas that diffuses

The physics here is more familiar than the exotic name suggests, it's the same idea as a drop of perfume spreading across a room, just applied to electrons instead of molecules. Treat the mobile charge carriers in a conductor as a gas. At the hot end, they carry more thermal energy and move faster; more of them wander from hot to cold than the reverse, for the same reason heat itself flows that direction. This is ordinary diffusion, nothing more exotic than what drives a gas to even out its density across a temperature gradient.

But unlike a neutral gas, these particles carry charge, so as they diffuse, they leave an imbalance behind: charge builds up on the cold side, and an electric field grows that pushes back against further diffusion. The system settles once that field is strong enough to turn away as many carriers as diffusion sends forward. The voltage you measure at open circuit is simply a readout of how strong that pushback field had to get.

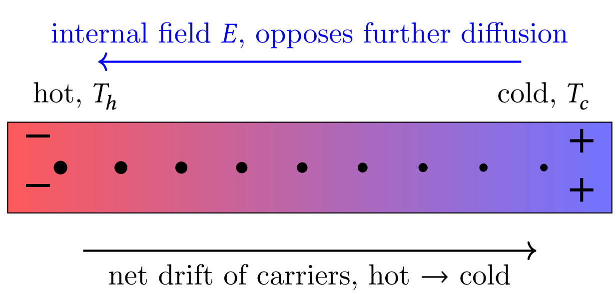


Figure 3: The Seebeck effect as ordinary diffusion of a charged gas. Faster carriers diffuse from hot to cold, charge separates, and the resulting field grows until it exactly cancels further net diffusion. The measured voltage is a direct readout of that balance.

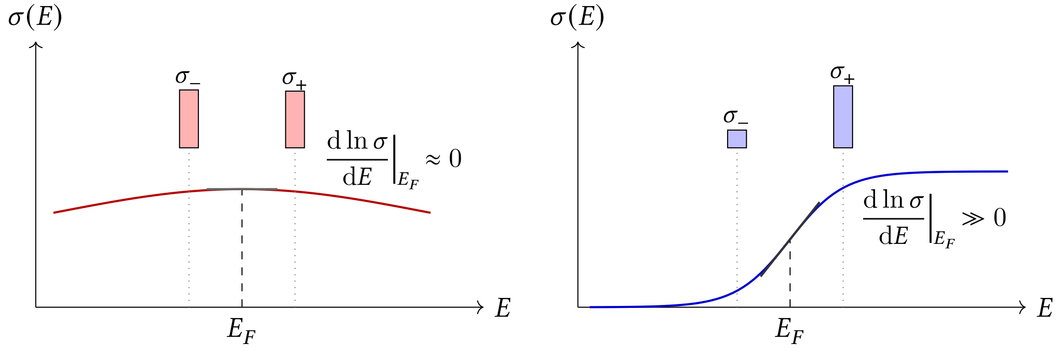
This qualitative picture becomes an exact formula through the Boltzmann transport equation [55], and for a metal or lightly doped semiconductor it reduces to the Mott formula,

$$S = \frac{\pi^2}{3}\frac{k_B^2 T}{e}\left.\frac{\mathrm{d}\ln\sigma(E)}{\mathrm{d}E}\right|_{E=E_F}. \tag{26}$$

It's worth pausing on what this formula is actually saying, because it overturns a natural but wrong intuition. You might expect S to be large whenever a material has lots of mobile carriers, more carriers, more current, bigger effect. That's wrong. Equation (26) says S depends on something else entirely: whether $\sigma(E)$, conductivity plotted as a function of carrier energy, looks the same on both sides of the Fermi level E_F, or not. Here's what "looks the same on both sides" means concretely. Pick a small step δ, and compare the conductivity a bit below E_F to the conductivity the same distance above it, $\sigma(E_F - \delta)$ against $\sigma(E_F + \delta)$. If a material's $\sigma(E)$ curve is shaped so those two values come out equal, the way a curve looks unchanged when reflected in a mirror standing at E_F, we call it symmetric there. If one side is bigger than the other, it's asymmetric, or lopsided. Equation (26) says S is a direct measure of exactly this lopsidedness, and nothing else. Think of it as a one-way turnstile: what matters isn't the size of the crowd, it's whether it's easier for carriers to squeeze through on the high-energy side of the gate than the low-energy side. A perfectly symmetric turnstile, our mirror case, lets equal numbers through in both directions no matter how big the crowd is, and no Seebeck voltage builds up. A lopsided one favors one direction, and a voltage builds up until the resulting electric field pushes back just hard enough to restore balance.

This is exactly why metals make lousy thermoelectrics. A metal has a huge number of carriers, but its conductivity is nearly flat, symmetric, across the Fermi level, so the turnstile is nearly balanced and S comes out to just a few μV/K. A lightly doped semiconductor is the opposite case: the Fermi level sits near a band edge, where conductivity is rising steeply on one side and nearly zero on the other, a very lopsided turnstile, and S can reach hundreds of μV/K. Good thermoelectrics, in other words, live in the awkward middle ground between metal and insulator, not at either extreme, which is exactly why the field is dominated by doped narrow-gap semiconductors like Bi_2Te_3 rather than by metals or true insulators.

That middle ground turns out to be even more awkward than S alone suggests. A good thermoelectric material needs a large S, but it also needs high electrical conductivity (so current actually flows) and low thermal conductivity (so the hot and cold ends don't just equalize by heat leaking straight through the material itself, short-circuiting the very gradient you're trying to exploit). These three demands pull in different directions, more carriers generally means more electrical conductivity but a smaller, more symmetric S, and



(a) Metal: symmetric gate $\Rightarrow$ small S (b) Semiconductor: lopsided gate $\Rightarrow$ large S

Figure 4: The Mott formula's "lopsided turnstile" made visual, with $\sigma_- \equiv \sigma(E_F - \delta)$ and $\sigma_+ \equiv \sigma(E_F + \delta)$. Dotted guide lines connect each bar down to the energy it was evaluated at. In a metal (a), the broad, smoothly varying density of states makes $\sigma(E)$ nearly flat near E_F: σ_- and σ_+ come out almost equal, the tangent line is essentially horizontal, and S is tiny. In a doped semiconductor (b), E_F sits on the steep flank near a band edge: σ_+ dwarfs σ_-, the tangent line is steep, and S is large. Rescaling either curve vertically, i.e. changing the total number of carriers, would change neither panel's conclusion, since Eq. (26) depends only on the curve's local shape at E_F, not its height.

materials that conduct electricity well usually conduct heat well too. The single (ZT) number engineers use to track this three-way tension is the thermoelectric figure of merit,

$$ZT = \frac{S^2\sigma}{\kappa} T, \tag{27}$$

and a large chunk of modern thermoelectric materials research is just the search for a material that breaks the usual rule linking σ and κ together [56]. The most successful strategy has a memorable name: phonon-glass electron-crystal (PGEC), engineer a material with a disordered, cage-like atomic structure that scatters heat-carrying vibrations (phonons) the way an amorphous glass does, while keeping a clean, ordered electronic structure that lets charge carriers move through almost unimpeded, the way a crystal does. You want heat to get lost and confused wandering through the lattice, while electrons sail straight through it.

7.1.2 From single materials to a usable device

To see why a single material isn't enough to build a usable device, return to the electron-diffusion picture. Take any short stretch of conductor with a small temperature difference across it, it doesn't matter whether that stretch belongs to the sample or to a length of connecting wire. Electrons near the hotter side diffuse toward the colder side faster than the reverse, so charge piles up on the cold side, building a local field that pushes back until diffusion and drift balance. This happens continuously, at every point along any wire with a gradient across it, each tiny segment settling into its own local balance, governed only by the local temperature there and the material's Seebeck coefficient S.

Stack these local contributions along a wire, and the total is the voltage across it. Because each segment's contribution depends only on its own local temperature change, and not on what came before or after, the running total depends only on where you started and ended, not on the path in between. That's the physical origin of path-independence: it isn't a mathematical trick, it falls directly out of every point along a homogeneous wire running the same local rule.

Now build the measurement. A voltmeter has two terminals, both part of the same instrument, sitting at one shared temperature T_0. Attach lead A to the sample's hot end and lead B to its cold end, and walk the full loop: out through lead A from T_0 to T_h, across the sample from T_h to T_c, back through lead B from T_c to T_0, arriving back where you started.

Here is the key fact: lead A, the sample, and lead B are all made of the same material. That means all three legs obey the identical rule converting a temperature change into a voltage, the same S, the whole way around. And because voltage is just that rule's output added up along a path, summing it around a loop that starts and ends at the same temperature, $T_0 \to T_h \to T_c \to T_0$, gives exactly what a direct trip from T_0 back to T_0 would give: zero. It's the same reason a hiker's net elevation change is zero over any round trip back to the same porch, gravity pulls the whole way, strong here, weak there, but height depends only on position, and returning to your starting position always nets to zero, regardless of the terrain in between.

Every local pile-up of charge along the path is completely real; the three legs simply arrange themselves so that no net charge separation is left between the meter's two terminals, because those terminals sit at the same starting and ending temperature. (This is also why superconducting leads fix the problem: a superconductor contributes essentially no voltage of its own, so the loop is no longer built from one consistent rule throughout, the cancellation breaks, and the meter reports the sample's true voltage. Absolute Seebeck coefficients are pinned down this way, or by comparison against a well-characterised reference material.)

The fix follows directly from the mechanism of the problem: a loop made of one material always nets to zero, so break that symmetry by making the outbound and return legs out of different materials, one with Seebeck coefficient S_1, the other with $S_2 \neq S_1$. Now the loop no longer traces one consistent rule all the way around, so the telescoping cancellation fails, and a real, nonzero voltage is left over between the meter's terminals. That's the fix, stated purely as a measurement trick: swap one lead's material to stop it from cancelling the other.

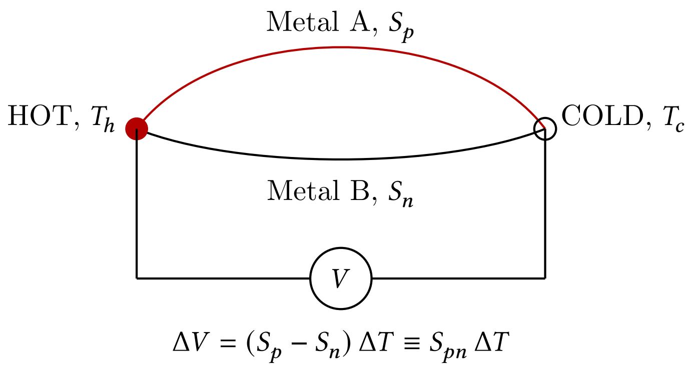


Figure 5: A single thermocouple, drawn as the classic two-junction loop. Metal A and Metal B connect the same two junctions, hot at T_h and cold at T_c, so both legs span the identical temperature drop. Because $S_p \neq S_n$, the loop is no longer built from one consistent rule, so the two legs' contributions no longer cancel, and the voltmeter reads their difference, $\Delta V = (S_p - S_n)\Delta T$, exactly Eq. (28).

But look at what that fix actually produced. Strip away the word "voltmeter" for a moment and look only at the physical object left behind: two dissimilar materials, joined at one end, sitting across a temperature difference, with a nonzero voltage appearing between their free ends. That description no longer contains anything about measuring, it's simply a device that turns a temperature difference into a voltage. Which is to say, it's already a generator. The same disagreement between materials that rescues a spoiled measurement is the only thing capable of powering a real load, and for the identical reason: a real difference has to be present around the loop, and it can only survive if the loop isn't built from one uniform rule.

So take that same idea and build it on purpose. Pair two dissimilar materials, conventionally an n-type leg and a p-type leg, joined at the hot end and left separate at the cold end, so the whole assembly forms a usable loop. The net open-circuit voltage of one such pair is

$$\Delta V = (S_p - S_n)\,\Delta T \equiv S_{pn}\,\Delta T, \tag{28}$$

and because $S_p > 0$ and $S_n < 0$ by construction, the two voltages add instead of cancelling, the entire point of choosing one leg of each type.

That's the reason every thermoelectric module you've ever seen, the credit-card-sized ceramic tile in a portable cooler or a CPU cooling block, is visibly built from alternating rows of differently coloured pellets: it's literally alternating p- and n-type legs, wired electrically in series and thermally in parallel, stacking Eq. (28) over dozens of couples until the output is a usable voltage instead of a few hundred microvolts.

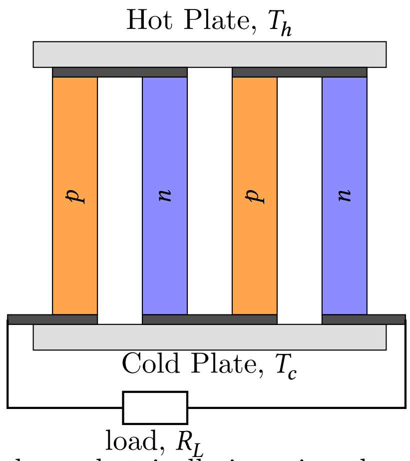


Figure 6: A thermoelectric module built from two p–n couples. Dark bars mark the copper straps: two join legs at the hot plate (making two p–n couples electrically in series) and one joins legs at the cold plate, so current threads up through one leg and down through the next all the way across the module. The two end legs, which have no partner at the cold plate, are the module's external terminals; their leads exit past the ceramic plate's edge and run down to the external load R_L, completing the circuit.

It's worth knowing that this effect has a mirror image. Run current into the same p-n couple instead of drawing it out, and one junction cools while the other heats, no compressor, no refrigerant, just electrons dragging heat along with them as they cross the junction. This is the Peltier effect, and it's not a coincidence that it uses the exact same device: Seebeck and Peltier are two faces of the same underlying transport process, linked by a simple exact relation (a special case of the Onsager reciprocal relations that also govern coupled transport elsewhere in this book), so any good thermoelectric generator material is automatically a good solid-state cooler, and vice versa.

And the Seebeck effect has been running, uninterrupted, for longer than almost any other piece of engineering in history: the radioisotope thermoelectric generators aboard the Voyager probes have been converting the heat of decaying plutonium into electricity via exactly Eq. (28), with no moving parts to wear out, since 1977, and are still faintly powering the spacecraft's instruments today, more than 24 billion kilometers from the sun.

7.2 The Peltier Effect

7.2.1 The reverse phenomenon

Thirteen years after Seebeck, Jean Peltier found the converse effect [47]. Driving a current I through a junction between two dissimilar conductors causes heat to be absorbed at one junction and released at the other, even in the absence of any temperature gradient and independent of ordinary I^2R (Joule) heating. The heat current carried through a junction per unit charge current defines the Peltier coefficient,

$$\dot{Q} = \Pi I, \tag{29}$$

with Π carrying units of volts. The sign of Π determines whether a given junction absorbs or releases heat for a given current direction, and reversing the current reverses which junction is hot and which is cold — this is precisely why a single thermoelectric module can be run as a cooler in one polarity and a heater in the other, with no moving parts to reconfigure.

The physical origin is again carrier diffusion, viewed from the opposite direction. Charge carriers do not just carry charge e; in a solid they also carry, on average, a kinetic and potential energy that depends on where they sit relative to the Fermi level and the local band structure. When carriers are driven across a junction between two materials with different average transport energy per carrier, the energy content of the current changes discontinuously at the interface. Energy conservation demands that the difference be dumped into, or pulled out of, the lattice as heat. This is the mechanism: current is forced across a junction where the carriers on either side disagree about how much energy per unit charge they are supposed to be carrying, and the junction settles the disagreement by absorbing or releasing heat.

7.2.2 The Kelvin relation: Peltier and Seebeck are the same effect

It would be a strange coincidence if Π and S were independent material properties, and indeed they are not. William Thomson (Lord Kelvin) [48] showed, using an argument built on the second law applied to the coupled charge and heat currents, that

$$\boxed{\Pi = ST} \tag{30}$$

This is one of the more satisfying results in the whole subject: it says the Seebeck and Peltier effects are not two effects that happen to occur in the same materials, but the same underlying coupling between charge and heat transport, observed under two different sets of boundary conditions (open circuit with a temperature gradient, versus isothermal with a driven current). Equation (30) is not an empirical fit; it is forced by thermodynamic consistency, in the same sense that the Maxwell relations of the previous section were forced by the exactness of a differential — here the analogous statement is Onsager reciprocity [49, 50] applied to the linear-response matrix relating charge and heat currents to their driving forces (electric field and temperature gradient), which is itself a microscopic consequence of time-reversal symmetry acting on the same second law.

7.3 The Thomson Effect and the Kelvin Relations

There is a third, smaller member of this family. William Thomson [48] also predicted, and later confirmed experimentally, that a homogeneous conductor carrying a current I while simultaneously supporting a temperature gradient will absorb or release heat continuously along its length, beyond ordinary Joule heating:

$$\dot{q} = \tau I \frac{\mathrm{d}T}{\mathrm{d}x}, \tag{31}$$

where τ is the Thomson coefficient. This effect requires neither a junction (unlike Peltier) nor an open circuit (unlike Seebeck) — it is the statement that even a single homogeneous material, carrying current through its own internal temperature gradient, redistributes heat. Thomson's second thermodynamic argument connects τ to the temperature dependence of S:

$$\boxed{\tau = T \frac{\mathrm{d}S}{\mathrm{d}T}} \tag{32}$$

Equations (30) and (32) together are called the Kelvin relations, and their practical importance is this: measure $S(T)$ for a material — a comparatively easy, local absolute temperature at the specific point, single-material, open-circuit measurement — and both Π and τ follow for free. Nobody needs to independently measure three coefficients for three effects; there is really only one transport property being probed three different ways, exactly as U, F, and G in the previous section were not three different energies but one relation viewed in three different natural variables.

7.4 Coupled Transport and the Origin of the Effects

It is worth writing the underlying transport equations explicitly, because they make clear that everything above is one linear-response relation, not three separate laws. In the presence of both an electric field $\mathbf{E}$ and a temperature gradient ∇T, the charge current density $\mathbf{J}$ and heat current density $\mathbf{J}_Q$ in an isotropic conductor are, to linear order,

$$\mathbf{J} = \sigma \mathbf{E} - \sigma S \, \nabla T, \tag{33}$$

$$\mathbf{J}_Q = \Pi \mathbf{J} - \kappa_e \, \nabla T, \tag{34}$$

where σ is the electrical conductivity and κ_e the electronic contribution to the thermal conductivity. Set $\mathbf{J} = 0$ (open circuit) in the first line and you recover the Seebeck effect, Eq. (25). Set $\nabla T = 0$ (isothermal) in the second line and you recover the Peltier effect, Eq. (29). The two off-diagonal coefficients that couple charge flow to heat flow and vice versa are related by the Kelvin relation, Eq. (30), which is nothing but the statement that the response matrix in Eq. (34) must be symmetric under the appropriate thermodynamic pairing of forces and fluxes — Onsager's theorem [49, 50] applied to this specific pair of currents. This is the same second law doing the same job it did throughout the potentials section: forbidding certain combinations of material response coefficients from being independent of one another.

7.5 The Figure of Merit and Why Most Materials Fail

7.5.1 Competing requirements

Knowing that thermoelectric conversion is possible says nothing about whether it is any good. A generator needs a large S (for voltage), a large σ (so the internal resistance does not eat the current you generate), and a small thermal conductivity κ (so the temperature difference driving the whole process is not short-circuited by ordinary heat conduction from hot side to cold side before it can be converted). These requirements are combined into the dimensionless thermoelectric figure of merit,

$$\boxed{ZT = \frac{S^2\sigma}{\kappa}T} \tag{35}$$

with $\kappa = \kappa_e + \kappa_L$ split into electronic and lattice (phonon) contributions [51, 52, 53, 54]. ZT is the single number that determines the maximum possible conversion efficiency of a thermoelectric device, in exactly the way that a Carnot efficiency bounds a heat engine — indeed the two limits connect directly:

$$\eta_{\max} = \underbrace{\frac{T_H - T_C}{T_H}}_{\eta_{\text{Carnot}}} \times \underbrace{\frac{\sqrt{1 + ZT_{\text{avg}}} - 1}{\sqrt{1 + ZT_{\text{avg}}} + T_C/T_H}}_{\text{thermoelectric reduction factor}}, \tag{36}$$

where ZT_{avg} is evaluated at the mean temperature. Equation (36) makes the physics unambiguous: a thermoelectric device cannot beat Carnot — nothing can — but for any finite ZT it falls well short of it, and only as $ZT \to \infty$ does the reduction factor approach unity. Present-day commercial materials cluster around $ZT \sim 1$, which for typical waste-heat temperature differences yields device efficiencies in the range of 5–8%. This is the reason thermoelectrics have historically lost out to mechanical heat engines wherever a mechanical engine is an option at all: not because the physics is exotic or unreliable, but because $ZT \sim 1$ simply is not a large number.

7.5.2 Why ZT is so hard to improve

The reason ZT has been stuck near 1 for the better part of a century, punctuated by slow, hard-won improvements [60], is that its ingredients fight each other through shared microscopic physics. Raise the carrier concentration to increase σ, and by the

Wiedemann–Franz law $\kappa_e = L\sigma T$ (with L the Lorenz number) you drag κ_e up in direct proportion, while simultaneously Eq. (26) tells you S tends to fall, because a higher carrier concentration pushes E_F further from the band edge and flattens the energy-dependence of $\sigma(E)$. Every knob you can turn to raise one ingredient of ZT tends to depress another, which is precisely why the field's biggest breakthroughs have not been about tuning carrier concentration harder, but about breaking the coupling itself.

Two strategies have dominated [59, 61]. The first is nanostructuring: introduce grain boundaries, nanoinclusions, or superlattice interfaces with length scales comparable to a phonon mean free path (tens of nanometres) but much smaller than an electron mean free path, so that lattice heat conduction κ_L is suppressed by boundary scattering while σ is left comparatively unharmed. This is the physical content behind the popular slogan "phonon-glass electron-crystal" — design a material that scatters heat-carrying phonons as effectively as an amorphous glass, while remaining as electrically conductive as a good crystal. The second is band engineering: convergence of multiple electronic valleys at the Fermi level (as exploited in modern PbTe and PbSe-based materials) increases the effective density of states available for conduction without the mobility penalty that simply adding more carriers would incur, partially decoupling S from σ. Together these two strategies are responsible for essentially most of the $ZT > 2$ results reported in the research literature over the last two decades, none of which yet dominate commercial modules, which remain built overwhelmingly on Bi_2Te_3-based alloys refined initially in the 1950s.

7.6 Device Operation: Generators and Coolers

7.6.1 The thermoelectric generator (TEG)

A thermoelectric generator is a p–n couple (or, practically, dozens to hundreds wired in series) sandwiched between a hot plate and a cold plate, with an external load resistance R_L connected across the terminals (see Figure 6). Electrically, the whole device is indistinguishable from a battery with some fixed internal resistance: a Thévenin source of open-circuit voltage $S_{pn}\Delta T$ and internal resistance R_{int} set by the leg geometry and material resistivity. The standard Thévenin result applies unchanged: load the source with anything other than $R_L = R_{\text{int}}$, and either too little current flows (large R_L) or too much voltage is lost internally (small R_L); matching the two resistances splits the available voltage evenly and maximizes the power actually delivered to the load,

$$P_{\max} = \frac{(S_{pn}\Delta T)^2}{4R_{\text{int}}}. \tag{37}$$

But matching resistances this way is also, by construction, wasteful: at $R_L = R_{\text{int}}$, exactly half the generated power is dissipated inside the device itself rather than delivered to the load, capping the best possible efficiency at 50% before any thermodynamics even enters the picture. This is the same trade-off as charging a battery through a resistor, maximum power delivery and maximum efficiency are simply different design targets, and the load resistance that achieves one is generally the wrong one for the other. A generator optimized for a spacecraft, where heat is essentially free and mass is everything, wants maximum power per unit mass and gladly loses that other 50%. A generator optimized for waste-heat recovery, where the heat itself has an opportunity cost, wants maximum efficiency instead, and is deliberately run away from $R_L = R_{\text{int}}$ to get it.

7.6.2 The Peltier cooler

Run current through the same device with no external heat source, and it becomes a solid-state heat pump. At the cold junction, three things happen simultaneously, and it is their competition that sets the achievable cooling:

$$\dot{Q}_C = \underbrace{S_{pn} I T_C}_{\text{Peltier cooling}} - \underbrace{\tfrac{1}{2} I^2 R_{\text{int}}}_{\text{Joule heating, half returns to cold side}} - \underbrace{K\,\Delta T}_{\text{parasitic conduction back}}, \tag{38}$$

where K is the thermal conductance of the legs in parallel. The Peltier term grows linearly with I and pumps heat away from the cold junction, exactly the mechanism from the Seebeck section run in reverse. The Joule term grows as I^2 and dumps heat right back into that same junction (only half of the total $I^2 R_{\text{int}}$, since by symmetry half flows to each junction). The conduction term is the most physically telling of the three: it isn't a fixed loss, it's a direct consequence of succeeding. Every kelvin of ΔT the device manages to create is a kelvin of ordinary thermal gradient sitting across the legs, and heat conducts back down that gradient exactly as it would through any material with a temperature difference across it, cold-junction pumping and hot-to-cold conduction leakage, in a race with each other, and the leak gets worse the better the pump is doing its job.

Because the Joule term is quadratic and the Peltier term is linear, there is necessarily an optimal current beyond which pushing more current through the device makes cooling worse, not better, a fact that catches out anyone who assumes a Peltier cooler will simply get colder if you turn up the power supply. And because the conduction leak grows with the very ΔT you're trying to achieve, cooling further becomes strictly harder the colder you already are, which is exactly why there's a hard ceiling on ΔT at all, not merely a best operating point. Differentiating Eq. (38) with respect to I gives the optimum current and, from it, the maximum achievable temperature difference at zero heat load,

$$\Delta T_{\max} = \frac{1}{2} Z T_C^2, \tag{39}$$

the direct, practical link between the microscopic figure of merit ZT from Eq. (35) and the number stamped on the datasheet of every Peltier module you can buy: a bigger ZT means the Peltier pumping term can outrun the Joule-and-conduction leak for longer as ΔT grows, pushing the ceiling higher. A typical commercial single-stage module with $ZT \approx 1$ near room temperature tops out around $\Delta T_{\text{max}} \approx 70$ K, which is why serious cryogenic cooling with Peltier elements requires cascading several stages, each one's cold side serving as the next stage's hot side, at rapidly diminishing efficiency.

7.7 Applications and Where the Physics Actually Gets Used

Radioisotope thermoelectric generators (RTGs) pair a radioactive heat source (historically $^{238}PuO_2$) with PbTe- or SiGe-based thermoelectric legs to power deep-space missions where solar power is impractical or, past the orbit of Mars, impossible. Voyager 1 and 2 have run continuously since 1977; their RTGs lose only a fraction of a percent of thermal output per year from the natural radioisotope decay, but lose thermoelectric conversion efficiency faster than that as ZT-degrading changes accumulate in the legs, which is the actual reason mission planners have had to progressively shut down instruments rather than any failure of the heat source itself.

Automotive and industrial waste-heat recovery targets the exhaust stream of combustion engines and industrial furnaces, where large ΔT and large, steady heat fluxes are available essentially for free. The barrier here has never been the physics — it has been that $ZT \sim 1$ materials, at automotive-relevant temperatures and with the mechanical and cost constraints of a production vehicle, recover only a few percent of the waste heat, and that has so far not cleared the bar against the added mass, cost, and complexity for the automotive industry, though the same devices are more favourably positioned in fixed industrial installations where mass is not a constraint.

Precision temperature control is, ounce for ounce, the application where thermoelectrics are least replaceable rather than merely competitive. Laser diodes, CCD and CMOS imaging sensors, IR detectors, and DNA sequencing instruments all require temperature stability at the millikelvin level, and a Peltier element's response speed, absence of moving parts and working fluid, and direct current-to-cooling-power relationship make it the default solution regardless of its modest efficiency, because in these applications efficiency is not the figure being optimised at all.

Wearable and IoT power harvesting exploits the small but nonzero ΔT between skin temperature and ambient air. The available power is tiny — typically tens of microwatts per square centimetre — but that is enough to run a low-power sensor node or a real-time watch display indefinitely without a battery, and it is a rapidly growing application precisely because the power budget of modern low-power electronics has fallen to meet what thermoelectrics can supply, rather than thermoelectrics improving to meet older power budgets.

The throughline across every application above is the same: thermoelectric devices win not when ZT is large in some absolute sense — it never has been, historically, compared to a mechanical heat engine's Carnot ceiling — but when the competing solution requires moving parts, a working fluid, scheduled maintenance, or simply does not fit in the available volume. The physics in this section, all of it inherited directly from the second law and Onsager reciprocity applied to coupled charge and heat transport, sets a hard ceiling on efficiency through Eq. (36). Everything that has happened in thermoelectric materials research for the last seventy years has been an argument about how close to that ceiling a real material, with real phonons and real electrons that refuse to be tuned independently, can actually get.

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